

An Engine of (Pay) Growth?

Productivity and Wages in the UK Auto Industry

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Abstract

To what extent do identical workers in an industry benefit from productivity growth in the same industry? We study this question in the context of the UK auto industry which experienced a thirteen-fold increase in output per worker between 1980-2018. Over the same period, auto workers saw substantially stronger hourly wage growth than workers in other manufacturing industries. Controlling for unobserved worker heterogeneity, the auto wage premium over other manufacturing workers doubled from 8% in the 1980s to 17% in the 2010s, while the manufacturing premium over the rest of the economy remained roughly stable at 5%. Interpreted through the lens of a rent sharing model, we estimate that 63% of the increase in the raw firm-level auto wage premium can be accounted for by strong productivity growth. In contrast, differential bargaining power between auto workers cannot account for these changes.

JEL classification: J3, L1, L6.

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1 Introduction

A long-standing tradition in labour economics has analysed why seemingly identical workers get paid a lot more, or a lot less, in some industries than in others. These inter-industry wage differentials have been found to be positively correlated with productivity. In the standard neoclassical model, productivity growth in one industry benefits workers indirectly because as consumers they will enjoy lower prices for the good or service produced. However, when labour markets are imperfectly competitive, wages in the same sector will generally adjust upwards to productivity shocks in that sector.

We contribute to this literature through an econometric case study of the UK auto industry.¹ This is an interesting context for at least three reasons. First, and perhaps most importantly, much of the literature has focused on why some industries *persistently* pay more than others. If this is so, then long-term changes in the structural features of an industry causing these wage differentials should also be reflected in pay. We argue that if the industry premium reflects quasi-rents relating to higher productivity in that industry, then a big increase in the sector’s relative productivity should be observed in a large increase in the sector’s relative wages. We document that an exceptionally large increase in productivity in the UK auto sector over the four decades from 1980 has gone hand-in-hand with similarly impressive growth in auto wages.

Second, the auto industry is often put forward as the poster child for a sector that gives good pay for blue collar workers. Krueger and Summers (1988) use autos as the example of an industry which has been a “wage leader” throughout the twentieth century (p.36). And this is sometimes used as justification for stronger industrial policy support through subsidies or tariffs, for example.

Third, there has been much recent work showing that the increase in individual wage inequality in the US and other countries has an important between firm component (Song et al. 2019; Faggio et al. 2010). Moreover, much of this firm premium appears to actually be growing differences between industries (Haltiwanger et al. 2024), which has re-awakened interest in inter-industry wage differences.

We document productivity trends in the auto industry in detail, drawing on data from firm-level surveys between 1980-2018. We show that the auto industry grew from just over 5% of manufacturing output in 1980 to over 11% in 2018. Output per worker rose by a factor of 13 compared to a factor of 4 in manufacturing as a whole and 2 in the market economy.² We show that much of this was due to the growth of global value chains (intermediate input intensity) and increases in capital intensity, but that there was also an impressive increase in Total Factor Productivity (TFP), which grew twice as fast as the manufacturing sector as a whole. UK auto productivity growth was faster than all other OECD countries where we could find comparable data.

Focusing on auto workers, we find the raw wage premium compared to the rest of manufacturing was 18 log points in the 1980s, but doubled to 37 log points in the 2010s.

1. We use the American “auto” and the British “car” interchangeably in this paper.

2. The market economy is the non-financial private sector of the economy, i.e. dropping government entities, health, education as well as finance, insurance and real estate.

Stripping out constant unobservable differences in worker quality reduces this premium to 8.2 log points (8.5%) in the 1980s but this composition-corrected value still doubled to 15.7 log points (17%) in the 2010s. In contrast, the manufacturing wage premium over the rest of the economy has been stable at around 5%.

We use a rent sharing model to analyse how much of the changing auto wage premium is due to increased productivity vs. other factors such as changing relative bargaining power. We estimate the rent sharing parameter and conclude that about 63% of the growth in the raw auto pay premium, without adjusting for worker composition, can be accounted for by the growth of relative productivity. We also find that worker bargaining power over rents has fallen over time, but this fall has been similar in autos and elsewhere in manufacturing. Hence, increasing relative bargaining power cannot explain the relative growth of auto wages.

Related literature

Our paper is related to several strands of the literature. First, we are most closely related to the work on inter-industry wage differentials, which focuses on why wages in some industries are higher than others, even after controlling for worker characteristics. Important contributions include Krueger and Summers (1988) and the seminal (“AKM”) Abowd et al. (1999) with an excellent early survey by Thaler (1989). Slichter (1950) pointed out the persistence of these differences in his 1923-46 data, with Card et al. (2023) reaching similar conclusions using data from more recent years. High paying sectors in one country tend also to be high paying in other countries, and the auto sector has some of the highest premia. We confirm this in our UK data, but also show that this wage premium has grown substantially over time and investigate the reasons for this change, linking it to productivity, which has been found to be an important correlate. This is important as there is no consensus as to the underlying causes of inter-industry wage differentials.³

Second, we contribute to the literature on the impact of firm-level characteristics on wages, which may be evidence of “rent sharing”. Numerous studies have documented evidence that firm effects are an important part of wage determination (Abowd et al. 1999; Card et al. 2018; Bonhomme et al. 2023). Moreover, positive shocks to firm performance are often reflected in higher employee pay, a phenomenon observed across various countries and datasets (Guiso et al. 2005; Carlsson et al. 2016; Friedrich et al. 2019).⁴

Third, there has been relatively little research on the *long-run* evolution of rent-

3. For example, whether they represent efficiency wages or unobserved worker ability - see Murphy and Topel (1990); Gibbons and Katz (1992); Dickens and Katz (1987). An older literature focuses on whether they represent compensating differentials due to non-pecuniary aspects of jobs.

4. There is substantial controversy over which is the appropriate imperfect competition model (Van Reenen 2024). Much recent work has focused on monopsony reasons (Manning 2011; Berger et al. 2022; Lamadon et al. 2022; Deb et al. 2023). An earlier literature focuses on bargaining rather than these wage posting models, either from individual search and matching (Mortensen and Pissarides 1994; Postel-Vinay and Robin 2002) or collective bargaining (Kalecki 1938; Leontief 1946; Van Reenen 1996). We are more in this tradition.

sharing. An exception is Bell et al. (2024), who construct a historical dataset of the top 300 British publicly listed firms finding that it has declined over time. However, it is an open question to what extent these findings generalise to other contexts over a larger representative sample of firms. Like them, we find a general decline in the rent sharing elasticity, which is consistent with falls in worker bargaining power. The change in autos is similar to other sectors in this respect. Furthermore, some papers have argued that the fall in bargaining power has led to a divergence between productivity and pay using US data (Stansbury and Summers 2020; Bivens and Mishel 2015) and Canadian data (Greenspon et al. 2021). We contribute to this literature by presenting evidence consistent with this trend occurring in the UK manufacturing sector.

Fourth, the decomposition of productivity into factor inputs and TFP and into between- and within-firm components is a further contribution of the paper (see *inter alia* Foster et al. (2001)). We find in autos almost all the change in productivity occurs within firms rather than between them, at least in the post-2000 period. We also find some evidence of the rise of superstar firms, however, which is a pattern observed in many contexts (Autor et al. 2020).

Finally, we contribute to understanding the auto industry, a sector that has been the focus of much work, especially in industrial economics (Berry et al. 1995). Unlike previous work, which tends to focus on issues of product market power, we make a novel contribution in examining the impact of productivity shocks on pay in this industry (we examine price-cost markups in the auto industry in more detail in a companion paper, Norris Keiller et al. 2024).

The paper proceeds as follows. Section 2 provides historical context of the UK auto manufacturing industry and section 3 provides a summary of our various data sources. Section 4 includes our analysis of productivity trends in the UK auto industry and overall manufacturing. Section 5 examines the evolution of wages using worker-level panel data before section 6 outlines a framework to examine the relationship between the trends documented in the two preceding sections. Section 7 concludes. Further details on the data and results are provided in online appendices A and B, respectively.

2 The UK Auto Industry in Historical Context

Before turning to the empirical analysis, we review key historical developments in the industry. There is a long tradition of car-building in Britain. Rover (now part of Jaguar Land Rover, JLR) was founded in 1878 and Rolls Royce in 1906. In 1955, the UK produced 1.2 million autos and was the world’s second largest auto manufacturer after the US. Decline set in, however, and the industry became a byword for inefficiency, industrial strife and poor quality. In 1968, the government engineered the creation of British Leyland, an ill-fated “national champion”, which was the merger of many separate manufacturers and ultimately BMC and Leyland Motors. Poor integration, weak management and industrial disputes continued. Despite (or because of) large government subsidies, however, the industry’s problems deepened in the 1970s, there was industrial turmoil (ten times as many days were lost to strikes in the UK than in Germany), with weak

management and falling sales. The election of Margaret Thatcher in 1979 prompted a cutback of public support, new management and a large reduction of the workforce. Since then, however, there has been a significant turnaround.

The fundamental causes of auto productivity growth have been the focus of a large literature. One important aspect has been the growth of foreign ownership (see Griffith 1999, for example). The government in the 1980s encouraged foreign multinationals to locate in the UK. In 1986, the Japanese auto firm Nissan entered Europe for the first time, building a manufacturing plant in Sunderland. The firm sought to engender the same commitment to quality, flexibility and teamwork as it had in its Japanese workforce with a single union agreement and binding arbitration, massively reducing industrial disputes. Similar strategies were sought in Toyota’s plant near Derby and Honda’s plant in Swindon. In 1994, the last of the large volume British owned auto manufacturers, Rover, was sold to Germany’s BMW.⁵ The growth of foreign ownership helped improve productivity directly, as foreign multinationals tend to have higher productivity (because of technological and managerial know-how - see Bloom et al. 2012) and indirectly through learning spillovers and higher competitive pressure (e.g. Alfaro and Chor 2023; Amiti et al. 2024).

In addition to FDI, being part of the European Single Market fostered stronger competition and opened new export markets (Owen 2010). The UK joined the European Union in 1975 and Thatcher (alongside Jacques Delors) helped build the Single Market in the early 1990s. This helped standardize regulations reducing the non-tariff (as well as tariff) barriers across a large swathe of manufacturing. Tariff barriers against European firms fell from 11% to zero after joining the EU. The larger market helped grow exports and provided competitive intensity. We show that the auto industry has started to decline again after the vote to leave the EU in 2016.

A third element of the wider institutional backdrop was an economy-wide decline in industrial disputes and union influence. Across the UK economy, the number of days lost to strikes fell from a peak of 27 million in 1984 to around 270,000 in 2018. This decline was not specific to autos and occurred in the economy more broadly.⁶

Finally, there is general technical change in auto manufacturing, alongside increased managerial know-how as the “Toyota Production System” of lean management diffused globally (Bloom and Van Reenen 2007). We show below that the improvements in the UK were particularly strong, suggesting that this is not the sole driver. Arguably, FDI, competition and better industrial relations helped speed up the spread of such technological and managerial know-how faster in the UK than elsewhere.

5. Note that the UK also has a thriving small volume high end “luxury” market (e.g. McLaren and Aston-Martin). Some of these remain UK owned (e.g. Morgan).

6. Figures taken from Office for National Statistics ‘Labour disputes; working days lost due to strike action’.

3 Data

To study the productivity and wage dynamics in the UK auto industry, we combine several data sources containing information on workers and firms. Taken together, these different datasets are a rich source of information on long-run developments in the industry. A full description is provided in Appendix A, but we provide an overview here.

On the firm side, we focus on data from representative surveys conducted by the Office for National Statistics (ONS): the Annual Business Survey (ABS, since 2008), the Annual Business Inquiry (ABI, 1998-2007) and the Annual Census of Production (ACP, 1980-1997). The key variables for our purpose are firms' nominal gross output,⁷ intermediate inputs, value added, employment and wage bill, which are collected on an annual basis. The sampling frame contains the universe of large firms (defined as having at least 250 employees) and a sample of smaller firms along with sampling weights to derive estimates which are representative of the population. We focus on the auto manufacturing industry, which we define through the industry code 'Manufacture of Motor Vehicles'.⁸ We also construct comparison samples for the whole of manufacturing and for the non-financial market economy. By merging the three surveys, we construct a consistent long-run sample which spans roughly four decades, 1980-2018. To put our results in a broader historical context, we also consider an extended sample which starts in 1973. However, a drawback of the extended sample is that the industry classification before 1980 does not allow us to distinguish between the manufacturing of autos and auto parts. We therefore start the main sample in 1980.⁹

On the worker side, we use the New Earnings Survey Panel Dataset (NESPD), which combines the Annual Survey of Hours and Earnings (ASHE) with its predecessor, the New Earnings Survey (NES), linking individual workers across datasets to facilitate panel analysis (Office For National Statistics 2022). The data contain information on key labour market variables such as earnings, hours and industry of employment for a 1% random sample of UK employees, selected by the last two digits of their National Insurance number (the UK equivalent of a US Social Security number assigned to all workers upon labour market entry). The data is reported by employers and covers both public and private sectors.¹⁰ Our analysis uses data from the period spanning 1982 to 2018. While the NESPD contains data from 1975, it is not possible to distinguish workers in the auto manufacturing industry from those working in auto part manufacturing until 1982 due to changes in the industry classification. Figure 13 in Appendix B shows auto part workers are consistently lower-paid, so this distinction matters.¹¹

7. For the purposes of this paper, nominal gross output is synonymous with revenue, sales and turnover.

8. The 'Manufacture of Motor Vehicles' industry category includes commercial in addition to personal vehicles. While this is clearly a distinct market, which may complicate the interpretation of our analysis, the majority of production in the sector is personal vehicles.

9. In addition, there is also a structural break in the wage variable.

10. It excludes the self-employed and employees paid less than the threshold for National Insurance payroll tax.

11. We end our sample period in 2018 as, although there is 2019 information in ASHE, we want to

4 Productivity trends in UK auto manufacturing

We start by documenting aggregate trends in the performance of UK auto manufacturers. We use firm-level data to show how the productivity of auto manufacturers (both labour productivity and TFP) evolved and compare these trends to those in the wider manufacturing sector.¹²

4.1 Industry-level productivity trends

Figure 1 shows the impressive productivity performance of the UK auto industry since 1980.¹³ In Panel A, we show real gross output (i.e. turnover) per worker, a measure of labour productivity. While we normalise productivity to 1 in 1980, the figure also shows the broader historical trend by including the period starting in 1973, which is the earliest year of the data.¹⁴

We see the problems discussed above in the 1970s and early 1980s, with productivity stagnating both in absolute terms and relative to the rest of the economy. Output per worker fell from £64,000 in 1973 to £60,000 in 1980 (these are deflated by an auto-producer price industry deflator and expressed in 2022 prices). Productivity starts to increase after this period in a secular way, peaking in the mid-2010s around the time of the Brexit referendum in 2016. At the end of the sample period in 2018, output per worker had risen by a factor of 13 compared to 1980, standing at £819,000 albeit with a sharp fall in the 2008-9 financial crisis. If we compare this to manufacturing as a whole over the same four decades, output per worker only rose by a factor of 4. In 1980 manufacturing productivity was £75,000 and rose to £312,000 in 2018. Thus, the auto industry had a labour productivity growth rate more than three times that of manufacturing as a whole. Labour productivity rose even more slowly in the wider market economy, continuing the usual trend of non-manufacturing having slower measured productivity growth (Baumol and Bowen 1965).¹⁵

be consistent with the ABI/ABS analysis used in the previous section. Moreover, there appears to be a large decline in the number of auto workers and their average wages (see Figure 14 in Appendix B), which is likely due to the ASHE survey week in 2019 coinciding with a period of temporary Easter closures in the industry.

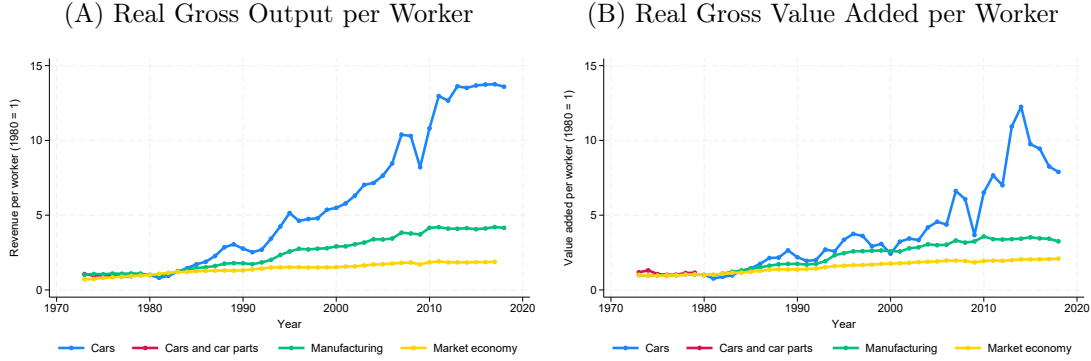
12. In the following section on worker outcomes we also use this data to examine changes in the labour share and relative wages.

13. The underlying values are reported in Table 9 of Appendix B.

14. We report the values prior to 1980 separately as the industry classification in this early period does not distinguish the manufacture of autos from the manufacture of auto parts.

15. The data for the market economy is taken from KLEMS which is derived from UK industry data, as the ABI/ABS does not contain information on the non-production sector before 1998.

Figure 1: Labour Productivity Growth



Note: panel (A) shows productivity in terms of gross output per worker for the auto industry, manufacturing and the market economy. Gross output is measured as turnover. Values are normalised to 1 in 1980. Panel (B) shows the identical figure for value added per worker.

One factor contributing to output growth is the increase in intermediate inputs. Outsourcing, especially internationally, was a key feature of the business landscape over this period and the auto industry is often the poster child for the globalisation of supply chains (Baldwin 2012). To investigate this, we plot an alternative measure of labour productivity, value added (VA) per worker, in Panel (B) of Figure 1. This strips out intermediate inputs from output. This series is a bit more volatile but still rose by a factor of 8 between 1980 and 2018 compared to a factor of 3 in manufacturing as a whole, and an even slower amount in the whole market economy. These auto improvements are especially pronounced after 2000. Hence, although the growth of outsourcing (mainly reflecting the growth of global supply chains) is an important part of the story of increased output per worker, it does not fully explain the overall growth, especially in the first two decades of the twenty-first century.¹⁶

Over 1980-2018, the share of auto in manufacturing gross output has risen from just over 5% to just under 12%, and the share of value added from 4% to just over 6%.¹⁷ By contrast, the share of jobs has remained stable over this period at around 3.5%. Hence, the growth of productivity has not resulted from a mass shedding of labour - at least compared to the rest of manufacturing.¹⁸

The auto series for gross output and value added in Figure 1 are deflated using auto-specific prices. We can also see productivity increases using estimates of the number of vehicles produced provided by the Society of Motor Manufacturers and Traders (SMMT). The raw quantity of vehicles per worker has increased four-fold (from 5.1 to 19.5).

16. In Figure 6 we show trends in the shares of intermediate inputs in nominal output that are consistent with these patterns. Intermediate shares were already much higher in autos than the rest of manufacturing in 1980, but rose strongly from under 60% to over 70% - about double the rate of other manufacturers.

17. This is also shown in Figure 7 in the appendix.

18. The auto value added share of GDP *has* fallen from 0.75% to 0.5%. But this is all due to the decline of manufacturing.

One reason for the strong growth in value added per worker is increases in capital intensity. To investigate this we calculate the growth in Total Factor Productivity (TFP). We measure TFP growth for industry k in year t , ΔTFP_{kt} , by a superlative index derived from the translog production (Caves et al. 1982; Griffith et al. 2004):

$$\begin{aligned} \Delta TFP_{kt} = & \Delta \log \left(\frac{VA_{kt}}{VA_{kt-1}} \right) - \frac{1}{2}(\alpha_{kt} + \alpha_{kt-1}) \Delta \log \left(\frac{E_{kt}}{E_{kt-1}} \right) \\ & - \left(1 - \frac{1}{2}(\alpha_{kt} + \alpha_{kt-1}) \right) \Delta \log \left(\frac{K_{kt}}{K_{kt-1}} \right) \end{aligned} \quad (1)$$

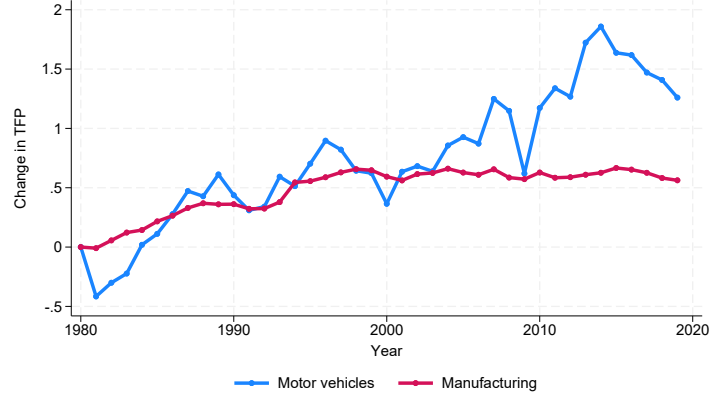
This formulation computes TFP growth as the difference between real value added (VA) growth and (weighted) labour (E) and capital (K) input growth, where the weight (α) is the labour cost share of nominal value added. We allow the labour share to vary over time and across industries, which is important as there has been a decline in the manufacturing labour share. To implement this decomposition, we draw on ONS estimates of productive capital stocks by industry.¹⁹

Figure 2 shows the trends in TFP in auto and manufacturing industries. The main result is that TFP has grown faster in the auto industry than in manufacturing. This indicates the stronger growth in labour productivity does not solely reflect increasing capital intensity. Auto TFP has risen by 1.2 compared to 0.5 in manufacturing as a whole, again with much of the growth happening after 2000.²⁰

19. We combine published numbers on productive capital stocks from two sources: one is a historical time series between 1980 and 2010 and the other is the most recent 'VICS' release which also covers the more recent time period. Note that due to differences in methodology over time, there is a level shift in these time series in 2010. We correct for this by adjusting the level of the post-2010 data so that it exactly matches the historical time series in 2010.

20. We also performed the decomposition using firm-level TFP instead of labour productivity and obtained very similar results.

Figure 2: Total Factor Productivity



Note: figure depicts TFP constructed using the index number approach of Caves et al (1982) for autos and manufacturing.

The strength of productivity growth in the auto industry is not common to all advanced economies. The UK turnaround stands out as a remarkable one, especially after many decades of relative decline. We use data from the OECD STAN database to compare the increase in productivity (real value added per worker using country-specific deflators) across countries, focusing on the US, Germany, France, Italy and Spain. Over the 1995-2018 period (for which the required data is available), productivity growth in the “Manufacture of motor vehicles, trailers and semi-trailers” sector was higher in the UK than in all other countries (see Appendix Table 10). UK real value added per worker growth was 170% compared to 140% in the US, 70% in France and Germany and 53% in Italy. The UK also had the fastest overall growth in auto value added of all these countries.

4.2 Firm-level Productivity analysis

To gain a deeper understanding of the factors driving productivity increases, we decompose productivity growth by firm, focusing on value added per worker at the firm level. Our main decomposition is a shift-share approach that distinguishes aggregate productivity growth due to within-firm growth, reallocation of employment shares towards more productive firms, and entry/exit (a more detailed description of this analysis is provided in Appendix B.2). Following Foster et al. (2001) we split aggregate productivity into that due to growth within firms vs. reallocation between survivors vs. net entry.

It turns out that productivity growth is almost entirely driven by within-firm productivity growth: reallocation of activity towards more productive firms and entry/exit only play a negligible role in the industry. The dominance of the within-firm component of productivity growth is different from most of the rest of the economy. This motivates the analysis of Section 6, where we develop a simple model of rent-sharing to examine

the effect of within-firm changes in productivity on wages.

We also examine whether within-firm improvements in productivity (and wages) have occurred evenly across the distribution or disproportionately among larger firms. To investigate this point, we split the auto and manufacturing samples into firm size terciles and compute average productivity for each tercile (see Appendix Figures 11 and 12). The key result is that the divergence in productivity (and wages) between the auto industry and manufacturing is primarily observed in the top tercile of the firm size distribution, with trends among smaller firms more similar to the rest of manufacturing.

4.3 Summary on auto productivity

Drawing on many datasets, we have documented a tremendous increase in real output per worker in the UK auto industry relative to the rest of manufacturing and the economy as a whole between 1980 and 2018. Increased outsourcing and capital intensity play an important role. Nonetheless, TFP has grown over twice as fast in autos as in total manufacturing. Auto productivity growth is primarily due to improvements within-firms, rather than between firms.

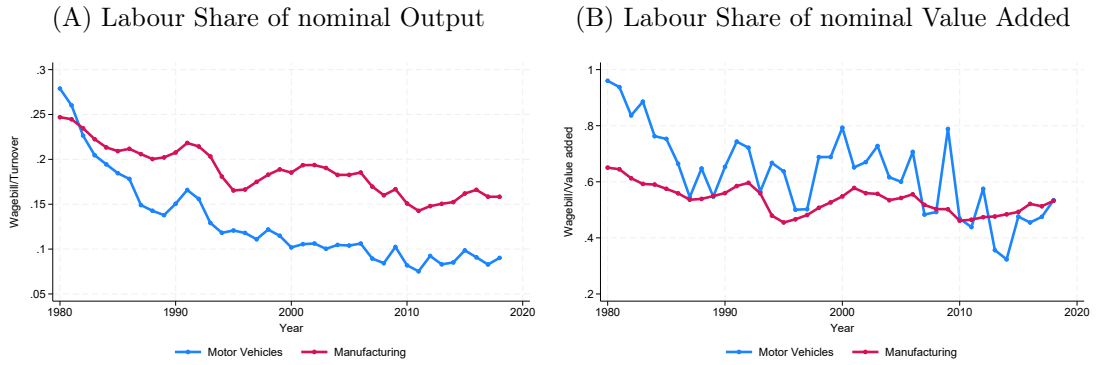
5 The Auto Wage Premium

5.1 Wage differences based on firm-level data

Figure 3 plots the labour share - the ratio of employment costs to nominal output (Panel (A)) and value added (Panel (B)) in the auto and manufacturing sectors.²¹ The labour share of output fell markedly in the 1980s and 1990s from 28% to about 10%, after which it remained relatively stable. It has declined even more in terms of value added: while labour captured nearly all of the auto industry's value added in 1980, by 2018 this had fallen to less than half. Labour shares in manufacturing have also declined since the 1980s, but less dramatically (see Teichgraeber and Van Reenen 2021).

21. Total employment costs include workers' compensation, national insurance contributions, pensions and redundancy costs.

Figure 3: The Labour Share



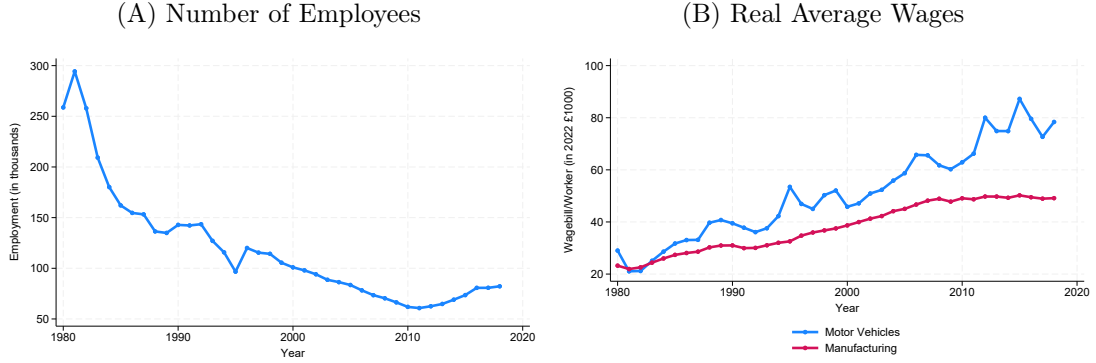
Note: panel (A) shows the labour share of nominal output (total employment costs divided by nominal gross output). Panel (B) shows the labour share of nominal value added.

A fall in the labour share does not necessarily imply a reduction in workers' wages. Indeed, Figure 4 shows the decline in auto industry labour share is exclusively due to reductions in the industry's workforce. Panel (A) shows that the number of employees fell from a quarter of a million in 1980 to 82,000 in 2018, with the bulk of this happening before the mid-1990s.

As shown in Panel (B), average wages measured by labour costs per worker actually pulled away from the rest of manufacturing. In the early 1980s, average wages were about £25,000 per year in both autos and manufacturing (in 2022 prices). Four decades later, average compensation had risen to about £78,000 in the auto industry compared to £49,000 in manufacturing more generally.²²

22. In Appendix Figure 12, we analyse heterogeneity along the firm size distribution by showing average wages for different firm size terciles. As with productivity, the divergence in average wages has primarily occurred among large firms.

Figure 4: Workforce and Average Wages



Note: panel (A) shows the number of employees in the auto manufacturing industry. Panel (B) shows aggregate wages (measured as firms' total employment costs) relative to the number of workers, deflated to 2022 prices using the CPI deflator.

5.2 Wage differences based on individual-level data

The results discussed above are based on firm and industry data, which abstracts from worker characteristics. One explanation of growing auto wage premium may simply be that the auto sector is becoming relatively more skilled. Additionally, auto workers may work increasingly longer hours, leading to a smaller *hourly* wage premium than that observed in the firm data, which is based on annual wages.

Table 1 contains summary statistics of our worker-level sample, overall and by decade. This shows average real wages across all sectors increased between the 1980s and the 2000s, but fell back in the 2010s due to the prolonged period of low wage growth following the global financial crisis. The workforce has become progressively older since the 1980s (an average of 38 in the 1980s compared to 41 in the 2010s), while the gender balance has become more equal (42% were women in the 1980s compared to 52% in the 2010s). Average working hours have steadily fallen (from 35 to 32 hours per week) and, in tandem, the prevalence of working part-time has risen (from 17% to 31%). The public sector's share of the workforce has fallen considerably from an average of 37% to 25%.

Table 1: Sample Summary Statistics (All Sectors)

	1980s	1990s	2000s	2010s	1982-2018
Age	38.07	38.56	40.03	40.83	39.43
Female (%)	0.42	0.46	0.50	0.52	0.48
Public (%)	0.37	0.29	0.26	0.25	0.29
Part time (%)	0.17	0.21	0.26	0.31	0.24
Weekly wage (£2022)	413	510	623	586	538
Basic hours	34.87	34.05	33.00	32.19	33.42
Hourly wage (£2022)	11.18	14.35	18.27	17.63	15.67
N weekly wage	1,296,533	1,585,909	1,531,875	1,576,280	5,990,597
N hourly wage	1,082,329	1,381,040	1,433,530	1,496,955	5,393,854

Note: table shows mean characteristics of workers in the NESPD by time period denoted in the column title (note “1980s” is 1982-1989, “1990s” is 1990-1999, “2000s” is 2000-2009 and “2010s” is 2010-2018). All statistics and sample sizes are unweighted.

We distinguish workers in the auto manufacturing and other manufacturing sectors based on the industrial code of their employer.²³ Table 2 shows average characteristics among workers in auto manufacturing and other manufacturing in the first and last decade of our sample period.²⁴ The average auto worker earned a slightly higher hourly and weekly wage than workers in other sectors in the 1980s, whereas in the 2010s they earned considerably more. For example, the increase in the hourly wage was £11.64 for auto workers compared to only £6.62 for other manufacturing workers. Looking at observable characteristics, aging in the auto industry has been a bit slower than in the rest of manufacturing, but the evolution of female workers and part-time work has been similar.

Tables 11 and 12 in Appendix B show additional summary statistics on occupation-specific employment and average wages by sector and decade. We distinguish between nine aggregate occupational categories that reflect the ‘Main Groups’ of the UK’s Standard Occupational Classification (SOC).²⁵ The auto manufacturing sector has experienced faster growth of professional workers (up by 5 percentage points more than the

23. The NESPD contains information on several different industrial classifications over different sample periods. From 1982 to 1995 the NESPD records information using the 1980 Standard Industrial Classification (SIC), from 1996 to 2008 the 1992 SIC and from 2009 onwards the 2007 SIC. Auto manufacturing workers as defined using industry code 3510/34100/29100 under the 1980/1992/2007 SIC. We define manufacturing workers using industry codes 2210-2959/15110-36639/10110-32990 under the 1980/1992/2007 SIC.

24. Table 12 in Appendix B shows the same statistics for the non-manufacturing private and public sectors.

25. The occupational classification recorded in the NESPD changes several times during our sample period in tandem with changes to the UK SOC taxonomy. We convert earlier classifications into the 2010 version of the UK SOC using a proportional mapping procedure that exploits dual-coded years to construct correspondence tables between classifications. This method is described in greater detail in the online appendix to Norris Keiller et al. (2020).

rest of manufacturing) and a faster upgrading from the least skilled group (elementary) to the next level (machine operatives). While the difference in average wages between auto and non-auto manufacturing has increased across all occupations, it has been largest among managers and non-customer services (the latter group is very small). Machine operatives stand out as the occupational category that has seen the biggest increase as a fraction of auto manufacturing employment rising from 20% in the 1980s to 40% in the 2010s. This occupational group has also seen a moderately large increase in average wages relative to the same occupational category in non-auto manufacturing.

Table 2: Summary Statistics by Manufacturing Sector

	1980s		2010s		Change		
	Cars	Other Manuf.	Cars	Other Manuf.	Cars	Other Manuf.	Cars-Other Difference
Age	40.98	38.42	42.86	43.03	1.88	4.61	-2.39
Female (%)	0.09	0.28	0.09	0.25	-0.00	-0.03	-0.03
Part time (%)	0.01	0.06	0.05	0.09	0.04	0.03	-0.01
Weekly wage (£2022)	527	445	968	673	442	228	214
Basic hours	38.57	37.54	37.10	37.19	-1.47	-0.35	-0.95
Hourly wage (£2022)	12.95	11.16	24.58	17.78	11.64	6.62	5.01
N weekly wage	9,617	350,890	5,749	134,839	15,366	485,729	501,095
N hourly wage	8,190	297,887	5,545	127,255	13,735	425,142	438,877

Note: table shows mean characteristics of workers in the NESPD by sector and time period denoted in the column title. The final three columns show the change in mean characteristics between the 1980s and the 2010s for each sector and in the difference between the auto sector and other manufacturing sector.

To quantify the auto wage premium controlling for worker characteristics, we estimate equations of the following form:

$$y_{it} = \beta_A AUTO_{it} + \beta_M MANUF_{it} + \beta X_{it} + \tau_t + \omega_w + \epsilon_{it} \quad (2)$$

where i denotes workers and t denotes years. y_{it} is either log weekly or log hourly wages. $AUTO_{it}$ and $MANUF_{it}$ are binary variables indicating whether a worker works in the auto industry and the general manufacturing sector respectively in year t . Note $MANUF_{it} = 1$ for workers employed in the auto industry in year t . Consequently, β_A reflects the auto wage premium over and above the manufacturing premium. X_{it} are worker characteristics including age, age squared, a public sector dummy, a part-time dummy, nine occupational dummies and indicators for whether wages are covered by collective bargaining.²⁶ τ_t are year dummies, ω_w are work area dummies and ϵ_{it}

26. We interact these collective bargaining dummies by period dummies to account for changes in the definition of collective bargaining in the NESPD.

is a mean-zero error.²⁷ We shall also allow for individual fixed effects in subsequent specifications.

Table 3 shows estimates of equation (2) for log hourly wages.²⁸ We present results by decade with and without controlling for worker attributes. The main coefficient of interest is the auto wage premium $\hat{\beta}_A$, given in the row titled ‘Auto manuf.’ We are particularly interested in changes over time. Columns (1)-(4) show results without controlling for worker attributes and (5)-(8) include controls.

In the first four columns, the raw hourly pay premium in manufacturing compared to non-manufacturing ($\hat{\beta}_M$) is about 3% to 5% and is stable over the four decade period. By contrast, there is an auto premium over the rest of manufacturing ($\hat{\beta}_A$) of 19 log points in the 1980s, which rises to 37 log points in the 2010s.

It is reassuring that the magnitude of the auto wage premium and its increase over time is not too dissimilar across decades to the results in Figure 4, which used firm-level data from an entirely different source (employment cost per worker at the firm level from the ABI/ABS). Including the observable controls in the last four columns shrinks the auto premium (and mildly increases the manufacturing premium to about 6%). The change is not dramatic, however - from 37 to 31 log points in the 2010s, for example - and the basic pattern of a large increase in the auto industry hourly wage premium is more pronounced (13 log points in the 1980s, rising to 31 log points in the 2010s).

Table 3: Hourly Wage Regressions by Decade

	(1) 1980s	(2) 1990s	(3) 2000s	(4) 2010s	(5) 1980s	(6) 1990s	(7) 2000s	(8) 2010s
Auto manuf.	0.188*** (0.007)	0.260*** (0.008)	0.326*** (0.010)	0.372*** (0.012)	0.131*** (0.006)	0.223*** (0.006)	0.275*** (0.008)	0.314*** (0.009)
Manuf.	0.036*** (0.002)	0.025*** (0.002)	0.046*** (0.003)	0.050*** (0.003)	0.059*** (0.002)	0.034*** (0.002)	0.058*** (0.002)	0.063*** (0.002)
N obs.	1,082,329	1,379,858	1,433,701	1,496,955	1,082,329	1,379,858	1,433,701	1,496,955
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Area FEs	No	No	No	No	Yes	Yes	Yes	Yes
Worker Characteristics	No	No	No	No	Yes	Yes	Yes	Yes
Worker FEs	No	No	No	No	No	No	No	No

Note: dependent variable is log real hourly wage. Worker characteristics include age, age squared and dummies for occupation groups, working part time and working in the public sector. Standard errors are clustered at the worker level and reported in parentheses. */**/** denotes significance at the 10/5/1% level respectively.

Although the auto wage premium is not explained by observable worker attributes,

27. ‘Work areas’ are recorded in the NESPD to reflect local labour markets. They are distinct from, and generally larger than, ‘travel to work areas’ (TTWAs), which are defined using census commuting data.

28. Table 14 in Appendix B shows the equivalent for weekly wages. We focus on hourly wage results, as these are the most appropriate outcome for gauging differences in the return to labour across sectors. While we include the weekly wage results for completeness, our findings are generally similar across wage measures.

it may still be the case that auto manufacturing workers differ in terms of unobserved quality. To control for unobservable worker heterogeneity, we therefore include worker fixed effects:

$$y_{it} = \beta_A AUTO_{it} + \beta_M MANUF_{it} + \beta X_{it} + \tau_t + \omega_w + \gamma_i + \epsilon_{it} \quad (3)$$

where γ_i are a full set of worker dummies and all other notation is the same as in equation (2).

Table 4 shows estimates of equation (3).²⁹ Again, we present results by decade both with and without worker controls, with the suite of worker controls now including worker fixed effects.³⁰ The inclusion of worker fixed effects means identification of the industry dummies is driven by workers who move between sectors (a point that we return to in subsection 5.3).

There is evidence that some of the auto worker premium is due to higher unobservable worker quality in autos as the premium shrinks by about half.³¹ Column (5), for example, indicates that the hourly wage premium for autos (β_A) in the 1980s was 8.5% (8.2 log points) compared to 14% (0.13 log points) in Table 3. Most importantly, however, we see the auto wage premium has grown strongly over time, rising from 8.5% (8.2 log points) in the 1980s to 17% (15.7 log points) by the 2010s. Hence, the bottom line is that the auto wage premium (over the rest of manufacturing) has broadly *doubled* over our four decade sample. This trend holds true regardless of whether we use the most rigorous controls or examine the raw differentials from individual worker or firm-level data.

29. The equivalent results for weekly wages are shown in Appendix Table 15.

30. The results in columns (1)-(4) without worker controls are different to the equivalent columns in Table 3, as a constant sample is imposed across columns in each table and the inclusion of worker fixed effects excludes workers who are only observed once in a decade. These workers will be included in Table 3 but excluded from Table 4. The results are extremely similar implying no sample selection bias from moving to this subsample.

31. Appendix Figure 15 plots average worker fixed effects by sector and decade. Workers in the auto sector consistently have higher estimated fixed effects, with the differential increasing across decades.

Table 4: Hourly Wage Regressions by Decade (Worker Fixed Effects Sample)

	(1) 1980s	(2) 1990s	(3) 2000s	(4) 2010s	(5) 1980s	(6) 1990s	(7) 2000s	(8) 2010s
Auto manuf.	0.181*** (0.007)	0.254*** (0.008)	0.323*** (0.010)	0.370*** (0.012)	0.082*** (0.008)	0.076*** (0.008)	0.113*** (0.013)	0.157*** (0.014)
Manuf.	0.033*** (0.002)	0.020*** (0.002)	0.042*** (0.003)	0.045*** (0.003)	0.051*** (0.002)	0.049*** (0.002)	0.072*** (0.002)	0.052*** (0.002)
N obs.	1,019,222	1,323,441	1,380,549	1,438,525	1,019,222	1,323,441	1,380,549	1,438,525
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Area FEs	No	No	No	No	Yes	Yes	Yes	Yes
Worker Characteristics	No	No	No	No	Yes	Yes	Yes	Yes
Worker FEs	No	No	No	No	Yes	Yes	Yes	Yes

Note: dependent variable is log real hourly wage. Worker characteristics include age, age squared and dummies for occupation groups, working part time and working in the public sector. Standard errors are clustered at the worker level and reported in parentheses. */**/** denotes significance at the 10/5/1% level respectively.

The inclusion of worker fixed effects in equation (3) means the auto premia are identified from switchers. The increase in this premium may be driven by an increase in the wage gain workers enjoy when joining the auto industry or an increase in the wage loss they suffer when leaving it. To gauge which of these is more pronounced, we estimate event studies of the form:

$$y_{it} = \sum_{l=-4}^4 \beta_l (\text{Lag } l)_{it} + \beta X_{it} + \tau_t + \gamma_i + \epsilon_{it}, \quad (4)$$

where $(\text{Lag } l)$ are lags and leads around the year in which worker i either joined or left the auto industry and all other notation is the same as in equation (3). We cap lags and leads at four years defining these variables as $(\text{Lag } l)_{it} = \mathbb{I}[t = \text{Event}_i + l]$ for $-3 \leq l \leq 3$, where Event_i is the year in which worker i joined/left the auto industry, as $(\text{Lag } l)_{it} = \mathbb{I}[t \leq \text{Event}_i + l]$ for $l = -4$ and as $(\text{Lag } l)_{it} = \mathbb{I}[t \geq \text{Event}_i + l]$ for $l = 4$.³²

When estimating equation (4) to quantify entry effects we restrict the estimation sample to workers who join the auto industry from another manufacturing industry and workers who are constantly employed in the auto industry. When quantifying exit effects, by contrast, we restrict the estimation sample to workers who leave the auto industry and move into another manufacturing industry and workers constantly employed in the auto industry. This means the ‘control group’, for whom l_{it} is set to the omitted -1 ‘base’ category, are consistently-employed auto manufacturing workers, although results are similar if we instead use consistently-employed non-auto manufacturing workers.

The interpretation of these event-study estimates relies on several standard assumptions. First, conditional on worker fixed effects, year effects, and observed controls,

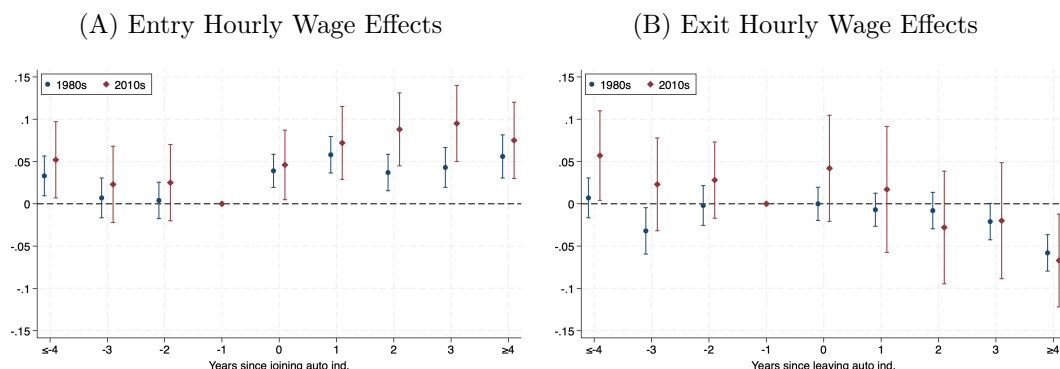
32. We omit area dummies from equation (4) for computational reasons. A considerable body of research that documents workers, particularly manufacturing workers, have limited geographical mobility (Autor et al. 2014).

transitions into and out of the auto industry are assumed to be exogenous with respect to transitory wage shocks. Second, we require a parallel trends assumption: absent entry into or exit from auto manufacturing, the wages of movers would have evolved similarly to those of workers who remain continuously employed in the auto industry. Finally, we require that a worker's wage path is unaffected by other workers' mobility decisions (i.e. the SUTVA assumption), which is plausible in this context given the focus on individual-level transitions within a large manufacturing sector.

Figure 5 plots the estimated lead and lag coefficients, $\hat{\beta}_l$, from separate entry and exit specifications using data from the 1980s and 2010s. The outcome variable in all specifications is log hourly wages, so the coefficient estimates can be interpreted as approximate percentage differences in wages relative to workers continuously employed in the auto industry. Across all panels, the coefficients on the pre-event leads tend to be insignificant, providing support for the parallel trends assumption underlying the event-study design.

Panel (A) of Figure 5 shows that a worker joining the auto industry from another manufacturing industry in the 1980s experienced an hourly wage increase of around 5 percent relative to continuously employed auto workers, with this difference remaining roughly constant following entry. While the initial wage gain for entrants in the 2010s is of similar magnitude, it continues to grow in subsequent years, reaching approximately 10 percent three years after entry.

Figure 5: Hourly Wage Event Study Analysis



Note: figures show point estimates and 95% confidence intervals for the entry/exit lead and lag variables of specification 4. Sample sizes in panel (A) are 9,069 for the 1980s and 5,749 for the 2010s. In panel (B) they are 14,708 for the 1980s and 4,386 for the 2010s. All panels additionally control for worker characteristics and worker and year fixed effects.

Panel (B) of Figure 5 presents similar estimates for workers that exit the auto industry. This shows wage declines among workers that leave the auto industry versus workers that remain only become significant four years after the exit transition. The estimated wage impacts of exit are comparable in the 1980s and the 2010s, with the magnitude of the loss similar across decades at around 6%. The relative similarity of the exit effects between the 1980s and the 2010s in comparison to the entry effects across decades suggests the increase in the auto wage premium documented above is more the result of increases in the wage gain of new entrants to the auto industry, rather than an increase

in the wage penalty of exiting the auto industry.

5.3 Calculating industry premia from firm wage premia

The industry wage premia in Table 4 are estimated using equation (3) and are hence identified from workers that move between industries. Card et al. (2023) argue that this identification strategy is likely to bias estimates of industry premia downward, as workers who newly enter an industry are likely to work in relatively low-paying firms compared to the industry average. They follow the seminal work of Abowd et al. (1999), who originally proposed calculating industry wage premia based on the weighted average of firm wage premia.

Following (Abowd et al. 1999), consider estimating firm fixed effects via an AKM wage equation of the form:

$$y_{it} = \beta X_{it} + \gamma_i + v_{j(i,t)} + \epsilon_{it}, \quad (5)$$

where $v_{j(i,t)}$ indicates that worker i worked for firm j at time t and all other notation is the same as in equation (3). Equipped with estimates of firm fixed effects $\hat{v}_j$, Card et al. (2023) suggest industry wage premia are calculated as differences in within-industry average firm fixed effects. For example, the analogue of the auto wage premia β_A of equation (3), which gives the auto manufacturing pay premium relative to manufacturing as a whole, would be calculated as:

$$\beta_A = \frac{1}{|AUTO|} \sum_{j \in AUTO} \hat{v}_j - \frac{1}{|MANUF|} \sum_{j \in MANUF} \hat{v}_j \quad (6)$$

where j denotes firms and *AUTO* and *MANUF* denote the set of firms in the auto industry and manufacturing sector respectively, meaning the $||$ operator denotes number of firms in each industry.

Estimating equation (5) requires observing the firm that a worker is employed in. This information is only available in the NESPD from 2004 onwards, which precludes a comprehensive comparison of the wage equation results above with their equivalents estimated via the Card et al. (2023) methodology. We instead focus on the latest decade of data, spanning 2010-2018 to gauge the magnitude of the bias highlighted by Card et al. (2023). As well as the comparison on the entire 2010-2018 sample, we consider a subsample of firms with at least 100 employees, as firm effects are relatively poorly identified among firms with few employees.

Table 5 summarises average firm fixed effect estimates by sector. Overall, the implied auto premia are comparable to those obtained from the wage regressions above. For example, in the “controls” version in the third column, the auto premium (relative to manufacturing) is 18 log points, which is close to the 15.7 log points in the final column of Table 4. If we use the large firm sample in the final column of Table 5, the auto premium is near identical to those in Table 4 at 16 log points.

Thus, while we cannot be certain that the comparison holds for earlier decades, these results lend confidence to our finding that the auto premium has increased. The

magnitude of bias that Card et al. (2023) highlight would have to be considerably greater in earlier decades than in 2010-2018 to overturn this result.

Table 5: Mean Firm Fixed Effects by Sector

	Raw		Controls	
	All	Emp. ≥ 100	All	Emp. ≥ 100
Non-manuf.	-0.02	-0.04	0.00	-0.02
Manuf.	0.10	0.09	0.08	0.07
Autos	0.41	0.39	0.26	0.23
Autos - Manuf.	0.32	0.29	0.18	0.16

Note: table shows mean firm fixed effects from a regression of log hourly wages by sector, weighted by employment. ‘Raw’ indicates firm fixed effects were estimated without controlling for worker characteristics or worker fixed effects. ‘Controls’ indicates firm fixed effects were estimated controlling for worker characteristics (quadratic age, part-time status, occupation, collective bargaining), and worker fixed effects.

5.4 Summary on trends in the auto wage premium

Combined with the results of the previous section showing that productivity increased in auto manufacturing relative to the rest of manufacturing, the rise in quality-corrected hourly wages is consistent with the rent-sharing hypothesis with which we began. This raises the question: is the increase in the auto wage premium solely due to increased productivity or has it also been influenced by changes in worker bargaining power? In the following section we use a simple framework to quantify the contribution of these factors but first we present robustness tests to corroborate our main findings.

6 The contribution of productivity to auto wage growth

6.1 Model

How much of the growth in the auto wage premium documented in section 5 can be attributed to the improved productivity in the auto sector documented in section 4? To answer this question, we draw on the following simple model of wage determination:

$$w_{ijt} = \alpha_{0t}^k + \alpha_{1t}^k p_{jt} + \alpha_t^k \eta_{ijt}, \quad (7)$$

where w_{ijt} is log wages of worker i in firm j in year t , p_{jt} is a measure of firm performance (e.g. log real value added per worker), and η_{ijt} are all other determinants of wages such as worker quality. The superscripts k denote industry, indicating that all parameters are industry-specific.

We focus on the wage premium in autos ($k = a$) compared to the rest of the manufacturing industry ($k = a'$). This can be denoted as:

$$\bar{w}_t^a - \bar{w}_t^{a'} = (\alpha_{0t}^a - \alpha_{0t}^{a'}) + (\alpha_{1t}^a \bar{p}_t^a - \alpha_{1t}^{a'} \bar{p}_t^{a'}) + (\alpha_t^a \bar{\eta}_t^a - \alpha_t^{a'} \bar{\eta}_t^{a'}), \quad (8)$$

where bars denote means calculated over workers and firms within an industry.

Adding and subtracting $\alpha_{1t}^{a'} \bar{p}_t^a$ from equation (8), we can decompose the auto wage premium into components due to productivity differences and bargaining differences:

$$\bar{w}_t^a - \bar{w}_t^{a'} = (\bar{p}_t^a - \bar{p}_t^{a'}) \alpha_{1t}^{a'} + (\alpha_{1t}^a - \alpha_{1t}^{a'}) \bar{p}_t^a + \zeta_t, \quad (9)$$

where the residual $\zeta_t = \alpha_{0t}^a - \alpha_{0t}^{a'} + \alpha_t^a \bar{\eta}_t^a - \alpha_t^{a'} \bar{\eta}_t^{a'}$ absorbs differences in the intercept and in the difference in other characteristics that affect wages across sectors. The leading term $(\bar{p}_t^a - \bar{p}_t^{a'}) \alpha_{1t}^{a'}$ is the contribution of the relative productivity difference in autos while we label the second term $(\alpha_{1t}^a - \alpha_{1t}^{a'}) \bar{p}_t^a$, as reflecting the contribution of differences in relative “bargaining power”.³³ In our baseline estimates, we evaluate the former at the non-auto bargaining parameter and the latter at the productivity level of autos but we also consider evaluating these terms at the auto parameter and non-auto productivity level respectively (and an average of the two) in robustness tests.

Differencing equation (9) across time we obtain:

$$\Delta(\bar{w}_t^a - \bar{w}_t^{a'}) = \Delta[(\bar{p}_t^a - \bar{p}_t^{a'}) \alpha_{1t}^{a'}] + \Delta[(\alpha_{1t}^a - \alpha_{1t}^{a'}) \bar{p}_t^a] + \Delta\zeta_t, \quad (10)$$

which we use to decompose changes in the auto premium $\Delta(\bar{w}_t^a - \bar{w}_t^{a'})$ into a component due to relative productivity changes $\Delta[(\bar{p}_t^a - \bar{p}_t^{a'}) \alpha_{1t}^{a'}]$, a component due to relative bargaining changes $\Delta[(\alpha_{1t}^a - \alpha_{1t}^{a'}) \bar{p}_t^a]$ and a residual component $\Delta\zeta_t$.

Implementing the decomposition in equation (10) requires estimates of the α parameters along with data on average wages and productivity. To obtain these quantities, we use the ABI/ABS data to estimate wage equations using the specification

$$w_{jt} = \beta_0 + \beta_1 \log\left(\frac{VA}{E}\right)_{jt} + \beta_2 \log\left(\frac{VA}{E}\right)_{jt} * AUTO_j + \phi_j + \tau_t + \epsilon_{jt}, \quad (11)$$

where j denotes firm and t denotes year.³⁴ $(\frac{VA}{E})_{jt}$ is value-added (VA) per employee (E) of firm j in year t and $AUTO_j$ is a dummy denoting whether a firm is in the auto industry. τ_t are time dummies and ϕ_j are firm fixed effects, which are included to flexibly control for the η terms of equation (8). Estimating this equation on a sample of manufacturing firms recovers estimates of the bargaining parameters of equation (8) as $\hat{\alpha}_1^{a'} = \hat{\beta}_1$ and $\hat{\alpha}_1^a = \hat{\beta}_1 + \hat{\beta}_2$.

Identification of the key parameters in equation (11) comes from variation in firm-level value added per worker over time. The baseline elasticity β_1 captures the relationship between wages and value added in manufacturing, while β_2 captures the differential elasticity for auto firms.

33. We use the term “bargaining” loosely as a shorthand for connecting wages to value added per worker and note the following points. First, the interpretation of the wage-productivity elasticity as a structural bargaining parameter will depend on the exact structure of the bargaining model. Second, the elasticity could be generated from a non-bargaining model, such as monopsony. Third, there are many econometric issues in interpreting the OLS elasticity as a causal effect. We discuss these issues in subsection 6.3.

34. In principle, one could estimate a worker-level analogue of equation (11). In our setting, however, this is not feasible because firm identifiers are available in the worker-level data only from 2004 onward, preventing us from linking workers to employer value added in earlier years.

Table 6 contains parameter estimates obtained by estimating equation (11) via OLS along with mean sample characteristics.³⁵ Estimates of the baseline elasticity for manufacturing ($\hat{\beta}_1$) imply that rent sharing has declined over time, falling from 0.27 in the 1980s to 0.18 in the 2010s. This is consistent with Bell et al. (2024) who find declines in rent sharing among large publicly listed British companies. We believe that we are the first to show this for a broader sample of firms that includes non-publicly listed firms. Stansbury and Summers (2020) argue that worker power to bargain over appropriate rents has also declined substantially in the US over this period.

The estimated differential elasticity for auto firms ($\hat{\beta}_2$) is not significant at conventional levels and close to zero in both time periods. Overall, the auto rent sharing term ($\hat{\alpha}_1^a = \hat{\beta}_1 + \hat{\beta}_2$) dropped from 0.24 to 0.14 over time, similar in magnitude to the drop in overall manufacturing ($\hat{\alpha}_1^{a'} = \hat{\beta}_1$). Hence, an increase in worker’s relative bargaining power does not appear to be a likely explanation for the rising auto wage premium.

Table 6: Auto Wage Premium Decomposition Estimates and Sample Moments

	(1) 1980s	(2) 2010s
Parameter estimates		
β_1	0.27 (0.011)	0.18 (0.018)
β_2	-0.03 (0.029)	-0.04 (0.024)
Sample mean characteristics		
Non-auto Log(Wage)	2.89	3.32
Auto Log(Wage)	3.04	3.79
Non-auto Log(VA/E)	3.17	3.92
Auto Log(VA/E)	2.99	4.74
Non-auto N	104,931	46,283
Auto N	340	335

Note: wages deflated using CPI, value added deflated using PPI. Parameter estimates and sample characteristics are calculated weighting by firm employment. Standard errors in parentheses clustered at the firm level. Sample sizes are unweighted. Underlying regression results are presented in Appendix Table 17.

6.2 Wage decomposition results

Table 7 shows results of the wage decomposition of equation (10), implemented using the estimates and sample characteristics of Table 6. An interesting implication is that the auto wage premium in the 1980s existed despite lower relative productivity: average value added per worker was lower in the auto industry than in other manufacturing, and

35. To facilitate comparison with previous analysis, results in the ‘1980s’ column are calculated using data over 1982-1989, while those in the ‘2010s’ column are calculated using data over 2010-2018.

both the productivity and measured bargaining components were negative. The positive premium must therefore instead be accounted for by the residual component (ζ_t). In the 2010s, by contrast, the positive gap between productivity in the auto industry and the rest of manufacturing accounted for about a third ($32\% = \frac{0.15}{0.47}$) of the auto wage premium. Between the 1980s and the 2010s, the raw auto wage premium grew by 32 log points (from 15 to 47). We calculate that 63% ($= \frac{0.2}{0.32}$) of this change is due to the faster growth of productivity in the auto industry. This was offset by differential worker bargaining power between the sectors. Without differences in bargaining power, the auto wage premium would have been approximately 38% ($= \frac{0.12}{0.32}$) greater than actually observed.

The residual component also increases substantially, from 0.29 log points in the 1980s to 0.53 in the 2010s, implying $\Delta\zeta_t = 0.24$. Because the decomposition uses raw firm-level average wages, ζ_t captures factors such as differences in worker composition and the sector-specific wage intercept. The worker-level results in Section 5 are consistent with part of this increase reflecting increasingly positive selection into the auto industry: the worker-fixed-effect-adjusted auto premium in Table 4 increases by only 0.075 log points, and Figure 15 shows increasingly positive worker fixed effects among auto workers. These estimates are not directly comparable because the firm and worker-level analyses use different wage measures. The 63% estimate above therefore refers to the share of the increase in the raw firm-level wage premium accounted for by productivity, rather than the composition-adjusted wage premium.

This decomposition evaluates the productivity contribution using the non-auto bargaining parameter and the bargaining contribution using the auto industry mean value added per worker. Table 21 in Appendix B repeats the decomposition of Table 7, but swaps the sector-specific inputs: the productivity contribution is now evaluated at the auto bargaining parameter, while the bargaining contribution is evaluated at the non-auto industry mean value added per worker. Under this decomposition, about 50% of the increase in the auto wage premium is due to changes in relative productivity, and the auto wage premium increase would have been approximately 25% greater had bargaining differences remained unchanged. Regardless of parameterisation, it is therefore clear that increases in auto industry productivity have significantly contributed to the widening auto wage premium and that differential bargaining power has not.

Table 7: Auto Wage Premium Decomposition

	(1)	(2)	(3)	(4)	(5)
		Decomposition components			
	$\bar{w}^a - \bar{w}^{a'}$	Productivity	Bargaining	Residual	N
1980s	0.15	-0.05	-0.09	0.29	105,271
2010s	0.47	0.15	-0.21	0.53	46,618
Change	0.32	0.20	-0.12	0.24	151,889

Note: column (1) shows the difference in log average wages between auto manufacturers and other manufacturers, defined as gross wages per employee. Columns (2)-(4) show the estimated components of equation 9. Column (5) shows the sample size used to calculate the wage difference and estimate the components.

This highlights that the magnitude of the productivity contribution to wages depends on the size of the bargaining power parameter. Our estimate of about 0.2 is in line with the pass through parameter of Friedrich et al. (2019), Kline et al. (2019) and Van Reenen (1996), although somewhat larger than those in the survey of Card et al. (2018). Using a lower bargaining parameter would reduce the share of the rising relative auto premium accounted for by rent-sharing.

6.3 Robustness of Wage Regressions

We implement several extensions to test the robustness of the wage regressions underlying the wage decomposition and summarise four of them here.

First, the estimates in Table 6 are obtained via OLS while there are many reasons why productivity may be endogenous. This is the subject of a vast literature (see e.g. Abowd and Lemieux (1993), Blanchflower et al. (1996), and Van Reenen (1996) and the survey in Card et al. (2018)), and in the absence of obvious external instruments we cannot do it justice here. We perform a simple exercise using lags as instruments as in the dynamic panel data literature using the moments suggested by Anderson and Hsiao (1981) and Arellano and Bond (1991). Specifically, we first-difference equation (11) and instrument the change in productivity (and its interaction with *AUTO*) with the lagged levels of these variables in $t - 2$ and $t - 3$. Under the assumption that ϵ_{jt} is serially uncorrelated, this approach yields consistent estimates of the β_1 and β_2 parameters. As shown in Appendix Table 19, the resulting IV estimates are qualitatively similar to the baseline results: the estimated elasticity of wages with respect to value added per worker declines between the 1980s and the 2010s, while the interaction is not statistically significant in either period.³⁶

Second, the estimates in Table 6 capture average relationships and may mask heterogeneity across firms. Prior research highlights substantial divergence between larger, more productive firms and smaller, less productive firms over recent decades (Autor

36. The linear elasticity term is larger than in Table 6, implying we would attribute a larger share of the growth in the auto wage premium to productivity.

et al. 2020). To explore whether similar heterogeneity is present in our setting, Appendix Table 20 reports estimates of equation (11) estimated separately for low- and high-productivity firms, where productivity is defined by comparing firms’ average value-added per worker in a given decade to the decade-specific median.

We find that the elasticity of wages with respect to value added per worker (β_1) declines between the 1980s and 2010s for both groups, but the magnitude of this decline is substantially larger among high-productivity firms. In contrast, we observe no statistically significant change in the coefficient on the interaction term (β_2) for either low- or high-productivity firms. Taken together, these results suggest that the weakening relationship between wages and firm performance is concentrated among more productive firms.

Third, there is an issue over what is the appropriate measure of the surplus that workers bargain over. Following the literature, we have summarised this as value added per worker, though this may be inappropriate if certain components of value added, such as quasi-rents from sunk capital, are excluded from bargaining. We follow Card et al. (2014) in augmenting equation (11) with a term in the log capital-labour ratio. Appendix Table 18 shows that the main patterns in Table 6 are robust to this additional control: the estimate of the rent sharing coefficient β_1 continues to decline between the 1980s and the 2010s, while the estimate of the bargaining coefficient β_2 remains statistically insignificant in both periods.

Finally, changes in worker sorting over time may also affect the relationship between wages and firm performance without reflecting changes in worker bargaining power. To the extent that higher-skilled workers increasingly select into high-productivity firms, part of the observed wage–value-added relationship may reflect changes in workforce composition rather than rent sharing *per se*. Firm fixed effects address time-invariant differences in worker composition across firms but do not eliminate time-varying sorting patterns.

The worker-level evidence presented in sub-section 5.2 provides useful context for assessing the importance of this channel. In particular, we document that the auto wage premium persists and increases over time even when worker fixed effects are included, and that average worker fixed effects in the auto sector rise over time. These results indicate increasing positive selection of workers into the auto industry. Such sorting would tend to strengthen, rather than weaken, the relationship between wages and firm performance, and therefore is unlikely to account for the declining wage–value-added elasticities documented in Table 6.

7 Conclusion

Economists have long been interested in inter-industry wage differentials. In this paper we use the UK auto industry as an econometric case study of whether a large persistent increase in industry productivity is reflected in industry wages for seemingly identical workers. We document a remarkable growth of labour productivity and TFP growth in the industry between 1980 and 2018.

Using panel data on workers and firms, we show that the wage premium in auto manufacturing has risen alongside the productivity boom. Even after controlling for worker fixed effects, auto workers in the 1980s enjoyed a wage premium of 8.5% compared to the rest of manufacturing, whereas by the 2010s this had doubled to 17%. Estimates from our rent sharing model imply that the majority of the increase in the raw auto pay premium can be accounted for by productivity increases (63% in the baseline case). At the same time, worker bargaining power seems to have declined in the auto industry as it has in the wider economy.

This suggests that the auto productivity boom was partially passed on to workers directly through higher wages, rather than solely through the effect of lower prices. We analyse prices and markups in the industry in a companion paper (Norris Keiller et al. 2024), where we find that markups for autos produced in the UK fell for the domestic market, but have risen for exports.

It is worth noting that productivity seems to have been declining in autos since around the time of the Brexit referendum in 2016. If this continues we predict that the pain will be felt not just in terms of jobs and profits, but also in terms of worker pay.

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A Data Appendix

A.1 ACP/ABI/ABS

In this Appendix, we describe the firm-level microdata in more detail. The data comes from annual surveys which have been conducted by the Office for National Statistics (ONS). The Annual Census of Production (ACP) started in 1970 and covered the production sector (manufacturing and mining). In 1998, the Annual Business Inquiry (ABI) was introduced, which contains data on the whole economy. The data between 1973-2007 is contained in Office For National Statistics (2012). In 2008, the ABI was replaced by the Annual Business Survey (ABS) (Office For National Statistics 2023). Note that even though the ACP starts in 1970, we focus on the period after 1980 as consistent time series can only be constructed for this period due to changes in variable definition in early years.

The surveys contain a sample of firms and provide sampling weights in order to produce representative statistics. The exact implementation of the sampling scheme has changed over the years. Broadly, the surveys contain the universe of large firms (with at least 250 employees) and a random sample of smaller firms, where the likelihood of being sampled depends on firm size and region.

In terms of industry definitions, we define our sample of the auto industry through the industry code “Manufacture of Motor Vehicles”. As the industry classification (UK SIC) has been adjusted over time, this refers to different industry codes:

- Code 3510 in the 1980 SIC
- Code 34100 in the 1992/2003 SIC
- Code 29100 in the 2007 UK SIC

Table 8 shows the number of observations in the auto industry and in the manufacturing sector for each year. Note that the data comes with sampling weights, as the firms in the data are typically larger than the population of all UK firms. Most notably, the drop in observations in the auto industry in 1979 is driven by a change in the sampling scheme

whereby coverage of smaller firms stopped being comprehensive and instead switched to a random sampling design.³⁷

Regarding variable definitions, we define value added (VA) as turnover minus intermediates. Note that the ONS in general has several alternative definitions of VA. The first is “value added at market prices”, which adds to our definition (turnover minus intermediates) the following categories:

- value of insurance claims
- change in value of stocks over the year
- net value of finished capital work for own use

The second is “value added at factor cost”, which further adds subsidies and net duty/excise payments and subtracts taxes relative to value added at market prices.

A final important consideration is the use of deflators to account for price changes over time. For the time period between 1996 and 2019, an industry-specific output price deflator time series is available from the ONS (the series "2910000000: Motor Vehicles"). For manufacturing, the ONS publishes a longer output price deflator series starting in 1957 (PPI series for “Manufactured Products for Domestic Market, Excl Duty”). In order to use the industry-specific information as much as possible while retaining coverage of the full period, we use the auto-specific deflator for 1996-2019. Before 1996, we assume the auto deflator changes with the same growth rate as the manufacturing deflator. The deflators are defined so real monetary values are expressed in 2022 prices.

37. All larger firms continued to be surveyed in each year of the survey.

Table 8: Number of Observations (ABI/ABS)

Year	Manufacturing	Motor vehicles
1973	20 354	434
1974	22 039	463
1975	20 409	416
1976	20 545	416
1977	20 472	402
1978	17 424	332
1979	17 138	43
1980	14 171	42
1981	14 206	45
1982	13 952	42
1983	13 572	45
1984	17 840	56
1985	13 302	42
1986	12 729	44
1987	12 807	47
1988	12 973	48
1989	18 452	49
1990	13 503	51
1991	13 323	55
1992	12 774	58
1993	11 362	55
1994	10 649	41
1995	9381	43
1996	10 554	54
1997	9147	45
1998	9139	49
1999	8936	51
2000	8739	48
2001	9159	51
2002	8507	49
2003	8277	47
2004	7970	44
2005	7500	41
2006	6780	42
2007	6988	47
2008	4298	44
2009	4016	38
2010	3946	32
2011	3668	33
2012	3780	31
2013	3652	30
2014	3748	34
2015	3667	35
2016	7693	49
2017	7458	50
2018	7049	48
2019	5664	45

Note: Table shows the number of observations in auto manufacturing and in manufacturing overall in our sample. The ABI/ABS provides sampling weights, so these observations represent a broader set of firms. The sampling scheme also changes over time, particularly in 1979.

A.2 NESPD

Our analysis of worker-level outcomes uses the New Earnings Survey Panel Dataset (NESPD), constructed by the UK ONS (Office For National Statistics 2022). The NESPD links the Annual Survey of Hours and Earnings (ASHE) to its predecessor, the New Earnings Survey (NES), creating a longitudinal dataset that tracks individual workers over time.

The NESPD covers a 1 percent random sample of employees in the UK, selected based on the last two digits of individuals' National Insurance numbers, which are assigned upon labour market entry. Earnings and employment information is reported by employers rather than workers, and the dataset contains detailed information on earnings, hours worked, and industry of employment. The data cover both the public and private sectors, but exclude the self-employed and employees earning below the National Insurance contribution threshold.

Although the NESPD is available from 1975 onward, our analysis uses data from 1982 to 2018. The start date reflects changes in industry classification that, prior to 1982, prevent reliable separation of workers in automotive manufacturing from those in automotive parts production. As documented in Appendix Figure 13, workers in auto parts manufacturing are consistently lower paid than those in vehicle assembly, making this distinction important for our analysis.

The sample period ends in 2018 to maintain consistency with the firm-level ABI/ABS data used elsewhere in the paper. While ASHE data are available for 2019, we exclude this year due to anomalous patterns in employment and wages in the automotive industry, likely related to the timing of the ASHE survey week coinciding with temporary Easter-related plant closures.

The NESPD contains information on several different industrial classifications over different sample periods. From 1982 to 1995 the NESPD records information using the 1980 Standard Industrial Classification (SIC), from 1996 to 2008 the 1992 SIC and from 2009 onwards the 2007 SIC. Auto manufacturing workers are defined using industry code 3510/34100/29100 under the 1980/1992/2007 SIC. We define manufacturing workers using industry codes 2210-2959/15110-36639/10110-32990 under the 1980/1992/2007 SIC.

B Further Results

B.1 Further descriptive evidence on productivity trends

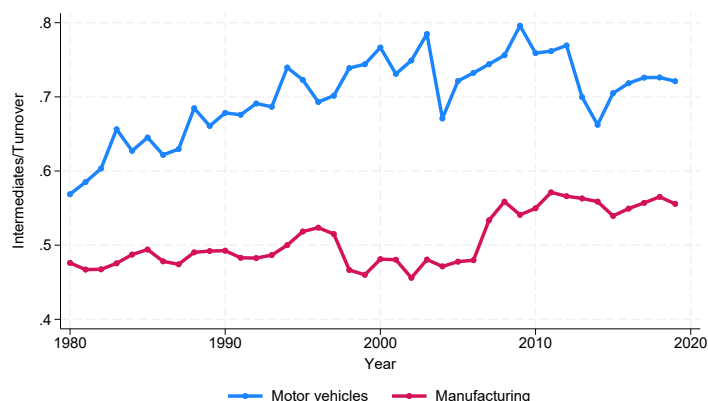
The main paper shows that value added per worker has risen more slowly than gross output per worker in autos, which is due to the increasing importance of intermediates. A more direct way to see this is through the rising ratio of the costs of intermediate inputs in total nominal gross output. We show this in Figure 6. The ratio was already much higher than the rest of manufacturing initially, but has also risen strongly since 1980 (from under 60% to about 70%), with the growth flattening off after 2000.

In terms of the broad trends Figure 7a shows auto manufacturing value added as a share of the UK's whole-economy GDP. The share is 0.5% in 2018, which is similar to the

level in 1990 and a decline relative to 0.75% in 1980.³⁸ This largely reflects the decline of manufacturing as a whole. Figure 7b shows that the output share of autos in overall manufacturing has risen from just over 5% to just under 12% over the four decades - more than a doubling. The growth of value added has been slower (consistent with Figure 1b), from just under 4% to just over 6%. By contrast, the share of jobs has remained stable over this period at around 3.5%. Hence, the growth of productivity has not resulted from a mass shedding of labour - at least compared to the rest of manufacturing.

To complement the value-based analysis of Figure 7, we also draw on industry estimates of the number of vehicles produced provided by the Society of Motor Manufacturers and Traders (SMMT), a trade organisation for the UK's motor industry, as a complementary, quantity-based way of analysing productivity. Figure 8b first shows that the quantity of vehicles produced in the UK has increased between 1980-2018 (from 1.3 million in 1980 to 1.6 million in 2018). The figure supports the notion that increases in the quantity of vehicles have contributed to productivity growth rather than being primarily driven by producing fewer but more expensive autos. Figure 8b also illustrates that this quantity-based measure of output per worker has increased four-fold (from 5.1 vehicles per worker in 1980 to 19.5 in 2018). This is smaller than the six-fold increase in value added per worker, but still substantial.

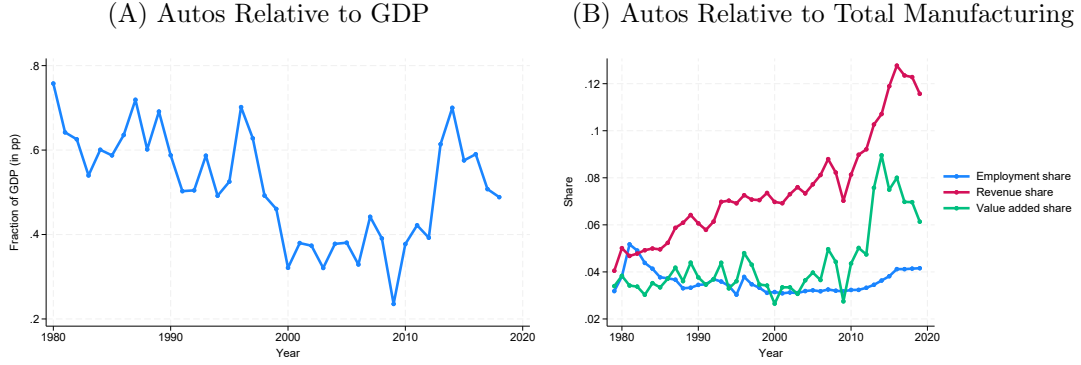
Figure 6: Share of Intermediate Inputs in Nominal Gross Output



Note: figure shows aggregate purchases of intermediate goods as a fraction of aggregate nominal gross output.

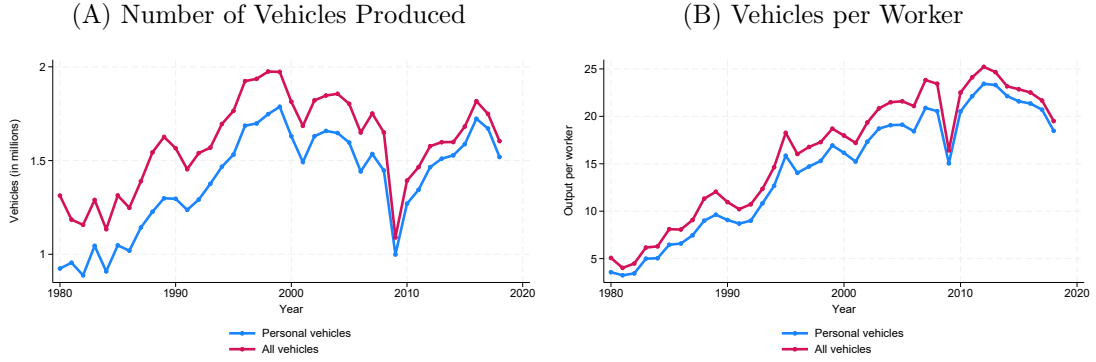
38. Unlike the 'market economy' series, GDP includes finance and the non-market sectors.

Figure 7: Relative Size of the Auto Manufacturing Industry



Note: panel (A) shows aggregate value added in the auto manufacturing industry as a fraction of GDP. Panel (B) plots aggregate output, employment and value added in auto manufacturing relative to the corresponding aggregate value in overall manufacturing.

Figure 8: Physical Output and Output per Worker



Note: panel (A) shows the number of vehicles (autos and commercial vehicles) produced by the UK auto industry. The data corresponds to industry estimates from the Society of Motor Manufacturers and Traders (SMMT), in particular from *The motor industry of Great Britain centenary book* (1980-1995) and the SMMT vehicle production press releases (1996-2018). Panel (B) shows the number of vehicles per worker using the industry estimates of output together with the employment information from the ABI/ABS.

B.2 Firm-level decompositions of productivity

To gain a deeper understanding of the factors driving productivity increases, we decompose productivity growth by firm, utilising data on value added per worker at the firm level. Our first decomposition is a shift-share approach that distinguishes aggregate productivity growth due to within-firm growth, reallocation of employment shares towards more productive firms, and entry/exit. We follow Foster et al. (2001) in using

Table 9: Labour Productivity in Autos and Manufacturing

Year	Revenue per worker		Value added per worker	
	Manufacturing	Cars	Manufacturing	Cars
1973	76.82	64.20	28.55	20.66
1974	80.63	57.53	27.56	22.97
1975	78.95	60.00	27.81	18.64
1976	82.44	56.06	27.66	17.62
1977	81.62	53.22	27.92	17.83
1978	83.02	55.89	29.28	19.85
1979	82.19	63.45	29.32	20.14
1980	75.35	60.27	28.61	17.51
1981	74.90	48.66	28.43	13.50
1982	81.71	56.82	31.29	15.38
1983	92.53	74.10	34.76	17.12
1984	102.39	87.85	37.02	22.40
1985	110.93	103.36	40.37	25.35
1986	114.22	113.58	43.26	30.47
1987	120.77	136.79	46.39	37.39
1988	132.13	171.96	49.12	37.89
1989	135.22	183.88	49.78	46.34
1990	134.66	166.55	49.95	38.37
1991	130.13	152.34	48.57	33.99
1992	138.96	162.32	49.96	35.07
1993	151.43	206.55	55.06	47.24
1994	176.47	255.83	66.56	45.30
1995	193.83	309.97	70.44	58.71
1996	207.20	278.32	73.94	65.64
1997	204.40	285.96	74.22	63.18
1998	207.91	288.83	74.95	51.12
1999	210.50	323.01	75.48	53.87
2000	219.31	330.85	74.14	42.52
2001	219.91	348.69	73.66	56.47
2002	230.12	379.82	79.64	60.19
2003	238.80	423.66	81.64	58.42
2004	255.28	431.14	87.25	73.20
2005	254.48	460.71	85.75	79.79
2006	259.08	509.93	86.43	76.62
2007	288.68	625.85	94.60	115.83
2008	284.61	620.52	90.59	106.35
2009	279.65	494.91	92.84	64.28
2010	312.84	651.03	102.32	114.02
2011	316.73	781.61	97.22	134.07
2012	308.96	763.01	96.42	122.59
2013	308.17	820.80	97.32	191.26
2014	311.42	814.30	98.06	214.34
2015	306.21	824.10	100.67	170.73
2016	309.91	827.72	98.76	165.29
2017	317.15	829.22	97.94	144.65
2018	312.82	819.08	93.05	138.17

Note: Table shows labour productivity values (output per worker and value added per worker) for each year since 1973. Values are expressed in 2022 GBP (thousands).

Table 10: International comparison of productivity growth (C29), 1995–2018

Country	Value added per worker
United Kingdom	170%
United States	140%
France	70%
Germany	70%
Spain	67%
Italy	53%

Note: This table shows growth in real value added per worker for the 1995-2018 period using data from the OECD STAN database. Value added is deflated using industry by country-specific deflators.

the decomposition:

$$\begin{aligned}
 \Delta z_t = & \underbrace{\sum_{i \in S} e_{it-1} \Delta z_{it}}_{\text{Within-firm}} + \underbrace{\sum_{i \in S} (z_{it-1} - z_{t-1}) \Delta e_{it} + \sum_{i \in S} \Delta z_{it} \Delta e_{it}}_{\text{Reallocation}} \\
 & + \underbrace{\sum_{i \in \text{Entry}} (z_{it} - z_{t-1}) e_{it} - \sum_{i \in \text{Exit}} (z_{it-1} - z_{t-1}) e_{it-1}}_{\text{Net entry}},
 \end{aligned} \tag{12}$$

where z_{it} is log value added per worker of firm i in year t , e_{it} is an employment weight (the share of jobs in the industry), and S indicates the set of firms that survived between year t and $t - 1$. z_t is the average productivity across all firms in year t .³⁹

Figure 9 shows the results from this exercise. Productivity growth is almost entirely driven by within-firm productivity growth: reallocation of activity towards more productive firms and entry/exit only play a negligible role in the industry.⁴⁰ The dominance of the within-firm component of productivity growth may seem surprising in the light of the reallocation literature which stresses between-firm components.⁴¹ This suggests

39. The decomposition is motivated by the fact that aggregate labour productivity can be rewritten as the employment-weighted sum of firm-level labour productivity (with VA_i being value added of firm i and E_i the number of employees):

$$\frac{\sum_i VA_i}{\sum_i E_i} = \sum_i \frac{VA_i}{E_i} \frac{E_i}{\sum E_i}.$$

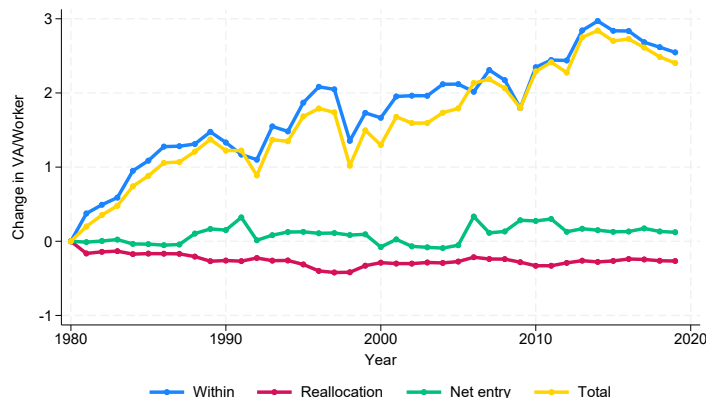
As a result, the decomposition directly corresponds to the earlier plot about aggregate productivity growth shown in Figure 1b.

40. In Appendix B we show that in manufacturing as a whole, net entry/exit plays a more important role. Note that since the data is a sample of firms rather than the population, the entry/exit component in the decomposition also picks up changes in sample composition, which is important to keep in mind in terms of the interpretation.

41. Note that the ABI/ABS samples firms at the level of *reference units*. This is a collection of establishments and a single enterprise can have multiple reference units. The decomposition in Figure 9 is performed at the level of reference units. For robustness, we also aggregated the data to the enterprise-level, which also shows the increase is driven by the within-component.

that the growth of productivity in the auto manufacturing sector comes from improvements within firms rather than employment becoming more concentrated in the most productive firms or changing employment shares across firms due to M&A and other competitive reshuffling of assets. We also performed the decomposition using firm-level TFP instead of labour productivity and obtain very similar results.⁴²

Figure 9: Shift-Share Decomposition of Labour Productivity Growth (VA/Worker)



Note: figure shows the results from a shift-share decomposition, which decomposes productivity growth into within-firm growth (Within-firm), reallocation of employment towards high-productivity firms (Reallocation), and entry/exit (Net entry). Total shows the overall productivity growth. See main text for definitions.

We also apply the Olley-Pakes (OP) decomposition which asks to what extent aggregate (employment-weighted) productivity is well-captured by the unweighted mean of productivity, as opposed to aggregate productivity being driven by large firms having an exceptionally high productivity. Formally, the OP decomposition expresses aggregate productivity as the sum of a mean and a covariance component:

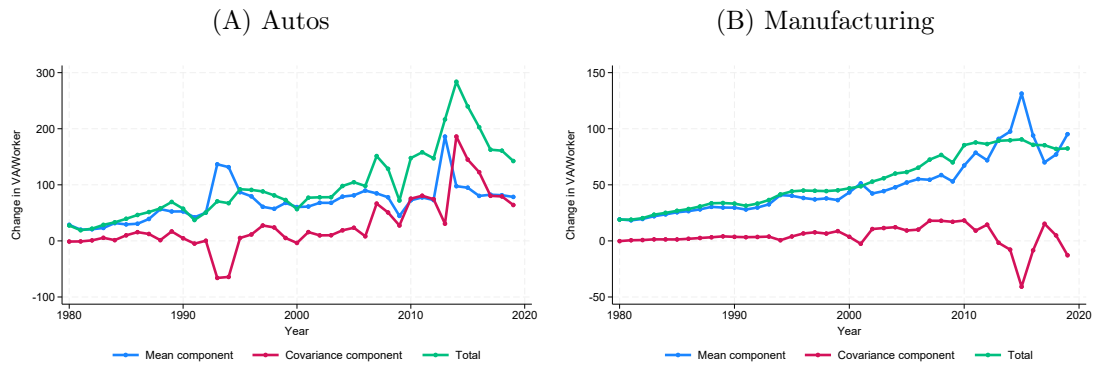
$$z_t = \sum_i w_{it} z_{it} = \underbrace{\frac{1}{N_t} \sum_{i=1}^{N_t} z_{it}}_{\text{Mean component}} + \underbrace{\sum_{i=1}^{N_t} (z_{it} - \bar{z}_t) (w_{it} - \frac{1}{N_t})}_{\text{Covariance component}}.$$

Like before, z_t/z_{it} are aggregate and firm-level productivity and w_{it} is the employment share. N_t is the number of firms in year t . Figure 10 shows the results. For the auto industry, aggregate productivity closely follows the unweighted mean until the late 2000s. Afterwards, the contribution of the covariance component becomes almost as significant as the mean component. This implies that the significant boom in value added per

42. In Appendix B, we also examine Olley-Pakes (OP) style static decompositions instead of the dynamic shift-share approach discussed above. The OP decompositions show that large firms play a significant role in driving the rise of productivity in the auto industry after 2010.

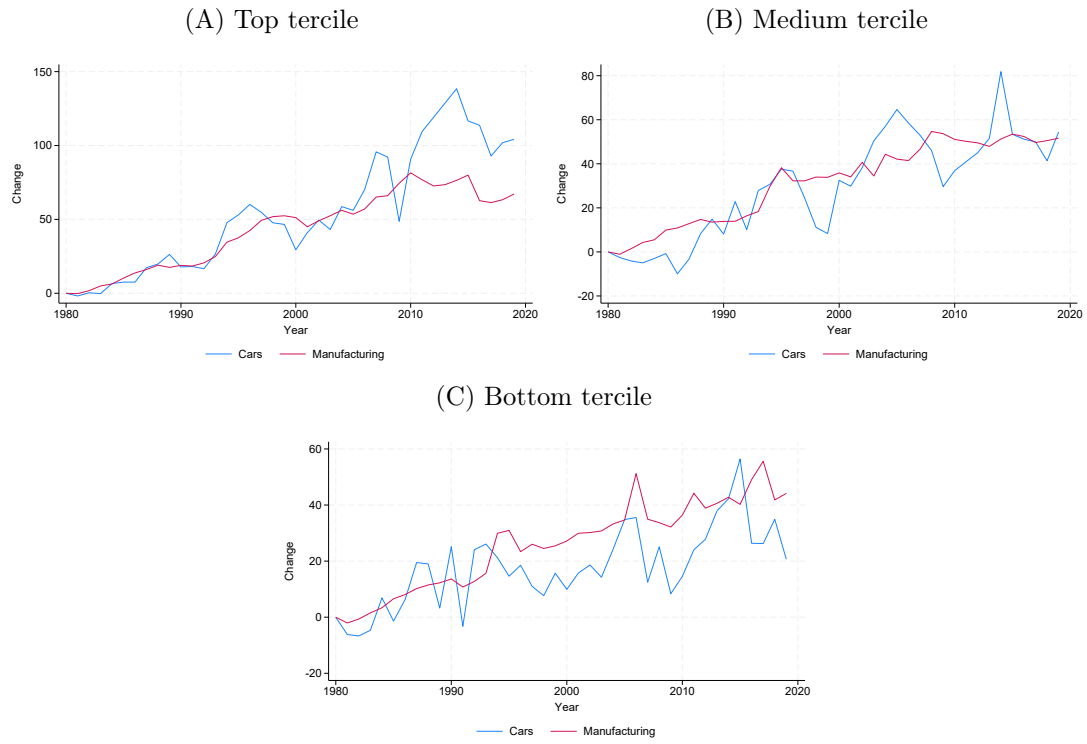
worker after 2010 is to an important extent driven by large firms who have very high productivity, which can be viewed as a “superstar firm” (Autor et al. (2020)) effect in the auto manufacturing industry. For manufacturing as a whole, by contrast, this superstar effect plays a smaller role and aggregate productivity growth is mostly close to the mean component.

Figure 10: Olley-Pakes Decompositions of Labour Productivity Growth (VA/Worker)



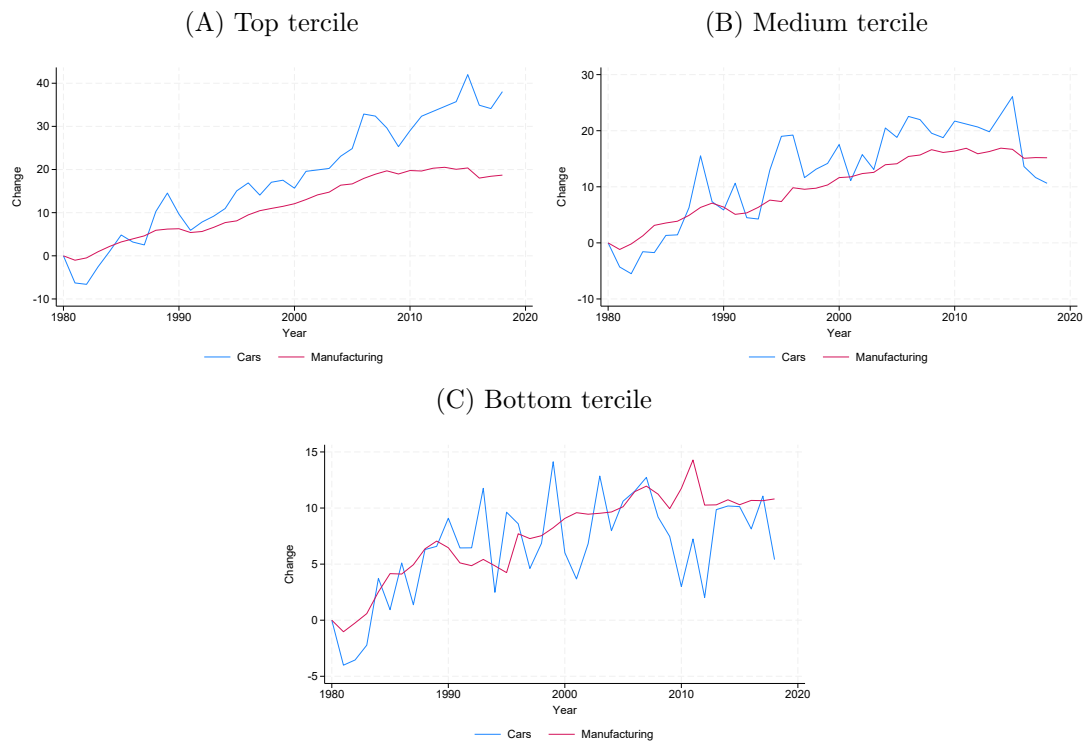
Note: figure shows the results from an Olley-Pakes decomposition of aggregate productivity (see main text for definition) for the auto manufacturing industry and manufacturing.

Figure 11: Productivity by Size Tercile



Note: figure shows average labour productivity (value added per worker) for different terciles of the firm size distribution, for the auto industry and manufacturing.

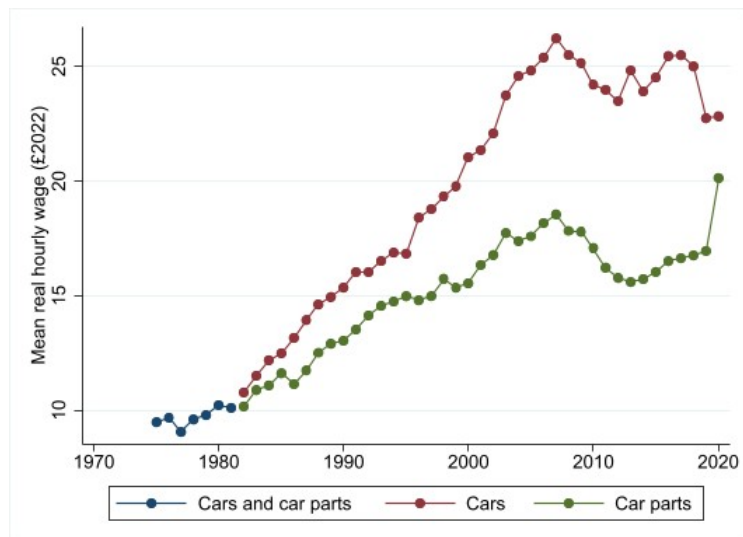
Figure 12: Wages by Size Tercile



Note: figure shows average wages (total employment costs over employment) for different terciles of the firm size distribution, for the auto industry and manufacturing.

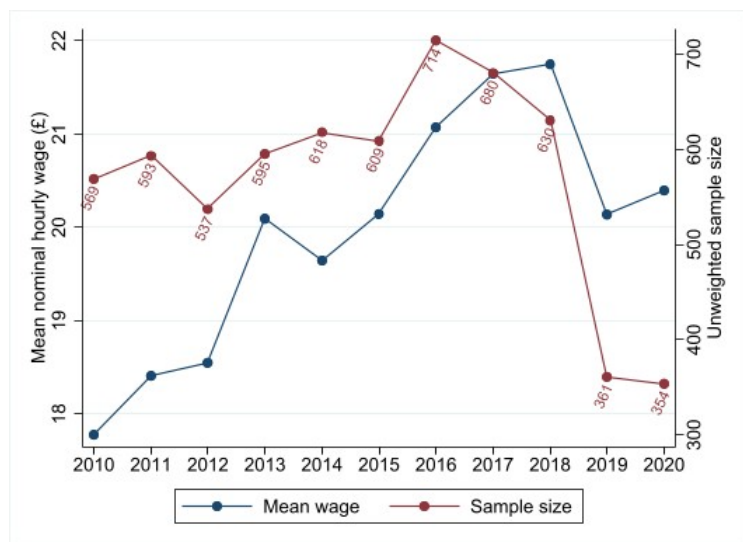
B.3 Further worker-level analysis

Figure 13: Mean Hourly Wage in Auto Manufacturing Industries



Note: figure shows average hourly wages in autos and auto parts based on the NESPD data.

Figure 14: Auto Manufacturing Industry Mean Hourly Wage and Sample Size



Note: sample size underlying the 'Mean wage' series is shown in the 'Sample size' series.

Table 11: Occupational Composition by Sector and Decade

	1980s		2010s		Change		
	Autos	Other Manuf.	Autos	Other Manuf.	Autos	Other Manuf.	Autos-Other Difference
Managers	0.06	0.07	0.07	0.09	0.01	0.02	0.01
Professional	0.10	0.09	0.15	0.10	0.06	0.01	0.05
Assoc. Pro.	0.11	0.11	0.09	0.13	-0.01	0.02	0.03
Admin	0.06	0.10	0.04	0.09	-0.02	-0.01	0.01
Skilled trades	0.31	0.23	0.19	0.18	-0.12	-0.05	-0.07
Other services	0.01	0.02	0.00	0.00	-0.01	-0.01	-0.00
Customer services	0.03	0.04	0.01	0.03	-0.01	-0.01	0.00
Machine ops	0.20	0.20	0.40	0.22	0.20	0.01	0.18
Elementary	0.12	0.13	0.04	0.15	-0.08	0.02	0.10
N	9,617	350,890	5,749	134,839	15,366	485,729	501,095

Note: table shows the share of workers in each major occupation group by sector and time period denoted in the column title. The final three columns show the change in occupation shares between the 1980s and the 2010s for each sector and the change in the difference between the auto sector and other manufacturing sector.

Table 12: Occupation Real Median Wages by Sector and Decade (2022 Prices)

	1980s		2010s		Change		
	Cars	Other Manuf.	Cars	Other Manuf.	Cars	Other Manuf.	Cars-Other Difference
Managers	15.85	13.69	35.80	27.06	19.95	13.37	6.57
Professional	15.82	15.22	31.17	27.67	15.35	12.45	2.91
Assoc. Pro.	14.78	12.92	25.98	22.07	11.20	9.15	2.05
Admin	10.97	11.37	17.00	15.10	6.03	3.73	1.50
Skilled trades	13.22	10.25	21.15	15.61	7.93	5.36	2.57
Other services	12.26	10.42	24.36	11.70	12.09	1.28	10.81
Customer services	12.83	10.89	19.37	14.48	6.54	3.58	2.95
Machine ops	12.07	9.11	21.07	13.41	9.00	4.30	4.70
Elementary	11.83	9.49	18.36	11.32	6.53	1.83	4.70

Note: table shows ‘fuzzy’ median wages by occupation, sector and decade calculated as the mean among the 50 observations closest to the median. The final three columns show the change in occupation ‘fuzzy’ median wages between the 1980s and the 2010s for each sector and in the difference between the auto sector and other manufacturing sector. Hourly wages are deflated using the Consumer Price Index and expressed in 2022 prices.

Table 13: Summary Statistics by Sector

	1980s		2010s		Change	
	Non-manuf.	Public	Non-manuf.	Public	Non-manuf.	Public
Age	36.10	39.80	39.56	43.49	3.46	3.69
Female (%)	0.43	0.52	0.50	0.69	0.07	0.17
Part time (%)	0.18	0.23	0.33	0.32	0.15	0.09
Weekly wage (£2022)	399	401	558	627	159	226
Basic hours	34.78	32.78	31.98	30.97	-2.80	-1.81
Hourly wage (£2022)	10.78	11.58	16.79	19.66	6.01	8.08
Occupational composition						
Managers (%)	0.10	0.06	0.09	0.04	-0.00	-0.02
Professional (%)	0.09	0.24	0.12	0.33	0.03	0.09
Assoc. Pro. (%)	0.13	0.11	0.11	0.15	-0.02	0.04
Admin (%)	0.20	0.14	0.14	0.18	-0.06	0.04
Skilled trades (%)	0.11	0.08	0.07	0.02	-0.05	-0.07
Other services (%)	0.04	0.10	0.09	0.17	0.05	0.07
Customer services (%)	0.11	0.05	0.16	0.02	0.05	-0.03
Machine ops (%)	0.09	0.07	0.06	0.01	-0.03	-0.05
Elementary (%)	0.13	0.15	0.16	0.08	0.03	-0.07
N weekly wage	477,375	458,651	1,050,340	385,352	1,527,715	844,003
N hourly wage	396,282	379,970	988,548	375,607	1,384,830	755,577

Note: table shows mean characteristics of workers in the NESPD by sector and time period denoted in the column title.

Table 14: Weekly Wage Regressions by Decade

	(1) 1980s	(2) 1990s	(3) 2000s	(4) 2010s	(5) 1980s	(6) 1990s	(7) 2000s	(8) 2010s
Auto manuf.	0.218*** (0.008)	0.289*** (0.009)	0.372*** (0.011)	0.429*** (0.013)	0.110*** (0.008)	0.190*** (0.007)	0.282*** (0.010)	0.296*** (0.011)
Manuf.	0.204*** (0.002)	0.236*** (0.003)	0.268*** (0.003)	0.299*** (0.004)	0.076*** (0.002)	0.049*** (0.002)	0.069*** (0.002)	0.077*** (0.002)
N obs.	1,296,533	1,584,673	1,531,867	1,576,280	1,296,533	1,584,673	1,531,867	1,576,280
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Area FEs	No	No	No	No	Yes	Yes	Yes	Yes
Worker Characteristics	No	No	No	No	Yes	Yes	Yes	Yes
Worker FEs	No	No	No	No	No	No	No	No

Note: dependent variable is log real weekly wage. Worker characteristics include age, age squared and dummies for occupation groups, working part time and working in the public sector. Standard errors are clustered at the worker level and reported in parentheses. */**/** denotes significance at the 10/5/1% level respectively.

Table 15: Weekly Wage Regressions with Worker Fixed Effects by Decade

	(1) 1980s	(2) 1990s	(3) 2000s	(4) 2010s	(5) 1980s	(6) 1990s	(7) 2000s	(8) 2010s
Auto manuf.	0.209*** (0.008)	0.283*** (0.009)	0.367*** (0.011)	0.424*** (0.013)	0.090*** (0.010)	0.079*** (0.010)	0.125*** (0.018)	0.151*** (0.019)
Manuf.	0.196*** (0.002)	0.227*** (0.003)	0.258*** (0.003)	0.288*** (0.004)	0.079*** (0.002)	0.076*** (0.002)	0.092*** (0.003)	0.085*** (0.004)
N obs.	1,237,024	1,528,321	1,480,848	1,520,217	1,237,024	1,528,321	1,480,848	1,520,217
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Area FEs	No	No	No	No	Yes	Yes	Yes	Yes
Worker Characteristics	No	No	No	No	Yes	Yes	Yes	Yes
Worker FEs	No	No	No	No	Yes	Yes	Yes	Yes

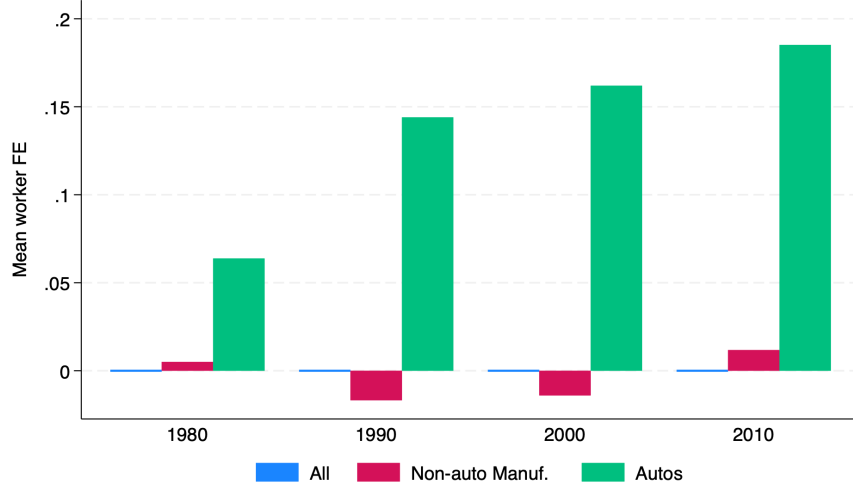
Note: see note to Table 14.

Table 16: Mean Weekly Wage Firm Fixed Effects by Sector

	Raw		Controls	
	All	Emp. ≥ 100	All	Emp. ≥ 100
Non-manuf.	-0.06	-0.06	-0.02	-0.03
Manuf.	0.34	0.35	0.15	0.15
Autos	0.70	0.67	0.34	0.30
Autos - Manuf.	0.36	0.32	0.18	0.16

Note: table shows mean firm fixed effects from a regression of log weekly wages by sector, weighted by employment.

Figure 15: Mean Worker Fixed Effects by Decade and Sector



Note: figure reports averages of worker fixed effects estimated using the specification in equation (3).

B.4 Wage Decomposition Regressions: Robustness

Our baseline implementation of the wage decomposition of equation (10) is carried out using the wage regression estimates given in Table 17.

Table 17: Auto Wage Premium Decomposition Underlying Regression Results

	(1)	(4)
	1980s	2010s
Log(VA/E)	0.271*** (0.011)	0.180*** (0.018)
Log(VA/E)*Auto	-0.029 (0.029)	-0.0440* (0.024)
N obs.	105,271	46,618
N firms	22,373	12,962

Note: standard errors in parentheses clustered at the firm level. */**/** denotes significance at the 10/5/1% level respectively.

As discussed in the main text, the trends in Table 17 may be driven by capital deepening. To examine this, we extend the baseline wage equation to include capital intensity. Because the capital stock is not directly observed in the ABI/ABS, we con-

struct firm-level capital using the perpetual inventory method based on the observed investment series. Table 18 reports the resulting estimates. For ease of comparison, columns (1) and (3) present the baseline specification without controls for capital per worker, estimated on the subsample of firms for which log capital is observed. Columns (2) and (4) then augment this specification by including log capital per worker as an additional regressor. Although the estimated coefficients corresponding to β_1 and β_2 — reported in the rows labeled $\text{Log}(\text{VA}/\text{Emp})$ and $\text{Log}(\text{VA}/\text{Emp}) * \text{Car}$, respectively — vary modestly across specifications (and capital intensity becomes significant in the 2010s), the key qualitative pattern is unchanged. The level of β_1 hardly falls when capital is included, and continues to decline between the 1980s and the 2010s. Similarly, β_2 shows no evidence of an increase. This indicates that our main conclusions appear to be robust to controlling for capital intensity.

Table 18: Auto Wage Premium Decomposition Regression - including capital intensity

	(1)	(2)	(3)	(4)
	1980s	1980s	2010s	2010s
$\text{Log}(\text{VA}/\text{E})$	0.269*** (0.012)	0.265*** (0.013)	0.180*** (0.018)	0.173*** (0.017)
$\text{Log}(\text{VA}/\text{E}) * \text{Auto}$	-0.0402 (0.028)	-0.0408 (0.028)	-0.0372 (0.024)	-0.0308 (0.022)
$\text{Log}(\text{K}/\text{E})$		0.00588 (0.005)		0.0819*** (0.010)
N obs.	79800	79800	40672	40672
N firms	16226	16226	11079	11079

Note: standard errors in parentheses clustered at the firm level. */**/** denotes significance at the 10/5/1% level respectively.

Our main estimates in Table 17 were obtained via OLS and could be biased if productivity is endogenous. To address this, we follow the dynamic panel data literature using an IV approach exploiting the moments suggested by Anderson and Hsiao (1981) and Arellano and Bond (1991). We first eliminate firm fixed effects by taking first-differences of equation (10) and then instrument $\Delta [\log(\text{VA}/\text{E})_{jt}]$ and $\Delta [\log(\text{VA}/\text{E})_{jt}] * \text{AUTO}_j$ with their lagged levels at $t - 2$ and $t - 3$. Specifically, we use the four instruments: $[\log(\text{VA}/\text{E})_{jt-2}]$, $[\log(\text{VA}/\text{E})_{jt-3}]$, $[\log(\text{VA}/\text{E})_{jt-2}] * \text{AUTO}_j$ and $[\log(\text{VA}/\text{E})_{jt-3}] * \text{AUTO}_j$. These are valid instruments under the assumption that the error term in the wage equation is serially uncorrelated.

Table 19 reports the results. Because the first-differenced instrumental variable (FDIV) estimation requires at least three consecutive observations per firm, we report our original OLS specifications (OLS with firm and year dummies) on the subsample for

which we can implement the FDIV estimator in columns (1) and (3). These firms are older and larger than the overall sample. Note that the coefficients are a little larger in this subsample than in the main sample, consistent with rent-sharing being more important for larger firms (e.g., Card et al. (2014)).

The comparable FDIV estimates are in columns (2) and (4) of Table 19. The F-statistics for the excluded IVs in the first stages are 13 or greater. The coefficient on the interactions of rent sharing with *AUTOs*, β_2 , continues to be insignificant as in the main results. The IV estimates of β_1 (shown in the row titled $\text{Log}(\text{VA}/\text{E})$) are larger for IV than OLS (0.47 vs. 0.33 for the 1980s and 0.26 vs. 0.19 for the 2010s), suggesting that we may be underestimating the importance of productivity in driving wage premia. Nevertheless, we continue to see a sizeable fall in the β_1 parameter over time of 0.21 (vs. 0.14 for OLS).

Table 19: Auto Wage Premium Decomposition Regression - IV estimation

	(1)	(2)	(3)	(4)
	1980s	1980s - IV	2010s	2010s - IV
$\text{Log}(\text{VA}/\text{E})$	0.330*** (0.022)		0.186*** (0.034)	
$\text{Log}(\text{VA}/\text{E}) * \text{Auto}$	-0.0202 (0.035)		-0.0410 (0.034)	
$\Delta \text{Log}(\text{VA}/\text{E})$		0.468*** (0.069)		0.256*** (0.065)
$\Delta \text{Log}(\text{VA}/\text{E}) * \text{Auto}$		0.0142 (0.145)		-0.0612 (0.175)
N obs.	31453	31453	11900	11900
N firms	7626	7626	2222	2222

Standard errors in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: standard errors in parentheses clustered at the firm level. */**/** denotes significance at the 10/5/1% level respectively.

Another exercise is to re-estimate the baseline model separately for low- and high-productivity firms. Firms are classified using a within-decade measure of productivity based on real value added per employee. For each decade, we compute the median level of real value added per employee and record the number of years in which each firm's productivity exceeds this threshold. Firms that are observed above the decade-specific median in more than 50% of their observed years within the decade are classified as high productivity, with the remaining firms classified as low productivity.

As noted in the main text, the auto sector interaction is not significantly different

from the rest of manufacturing at the 5% level (if anything, there is somewhat less rent sharing, especially in the 2010s for the low productivity firms). The fall in the rent sharing coefficient is strongest for the high productivity firms.

Table 20: Auto Wage Premium Decomposition Regression - by firm productivity

	(1)	(2)	(3)	(4)
	1980s low prod.	2010s low prod.	1980s high prod.	2010s high prod.
Log(VA/E)	0.243*** (0.015)	0.208*** (0.035)	0.287*** (0.014)	0.160*** (0.015)
Log(VA/E)*Auto	-0.0149 (0.046)	-0.0627* (0.033)	-0.0321 (0.034)	-0.0281 (0.025)
N obs.	48058	21235	57213	25383
N firms	10527	6146	11846	6816

Standard errors in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: ‘low prod.’ denotes low-productivity firms. Firms are classified as low productivity if their average productivity in a given decade is below the decade-specific median level. Conversely, high-productivity firms denoted by ‘high prod.’ are those with above-median productivity. Standard errors in parentheses clustered at the firm level. */**/** denotes significance at the 10/5/1% level respectively.

Table 21: Auto Wage Premium Decomposition Alternative Parameterisation

	(1)	(2)	(3)	(4)
	$\bar{w}^a - \bar{w}^{a'}$	Portion due to differences in		
		Productivity	Bargaining	N
1980s	0.15	-0.04	-0.09	105,271
2010s	0.47	0.11	-0.17	46,618
Change	0.32	0.15	-0.08	151,889

Note: column (1) shows the difference in log average wages between auto manufacturers and other manufacturers, defined as gross wages per employee. Columns (2)-(3) show the estimated components of equation 9. Column (4) shows the sample size used to calculate the wage difference and estimate the components.